Chapter # 01
Digestive System of Man

Student Learning Outcomes

After studying this chapter, students will be able to:
- [B-12-R-24] Describe the mechanical and chemical digestion in the oral cavity.
- [B-12-R-25] Explain swallowing and peristalsis.
- [B-12-R-26] Illustrate with a diagram the structure of the stomach and relate each component with mechanical and chemical digestion in the stomach.
- [B-12-R-27] Identify the role of the nervous system and gastrin hormone on the secretion of gastric juice.
- [B-12-R-28] Describe major actions carried out on food in the three regions of small intestine.
- [B-12-R-29] Trace the absorption of digested products from the small intestine lumen to the blood capillaries and lacteals of the villi.
- [B-12-R-30] Describe the component parts of the large intestine with their respective roles.
- [B-12-R-31] Correlate the involuntary reflex for egestion in infants and the voluntary control in adults.
- [B-12-R-32] Explain the storage and metabolic role of liver.
- [B-12-R-33] Describe the composition of bile and relate the constituents with respective roles.
- [B-12-R-34] Outline the structure of pancreas and explain its function as an exocrine gland.
- [B-12-R-35] Relate secretion of bile and pancreatic juice with the secretin of hormone.

The human body is a complex machine that requires a continuous supply of energy to perform countless biological processes. This energy comes from the food we eat - but before it can be used by our cells, the food must be broken down into simpler, absorbable components. This intricate process is carried out by the digestive system.

The digestive system is not just a group of organs working in isolation. It is a highly coordinated system that involves mechanical and chemical digestion, neural and hormonal control, and absorption and assimilation of nutrients. Starting from the oral cavity and ending at the anus, each part of the digestive tract performs specific roles to ensure that nutrients are extracted efficiently and waste is removed from the body.

SLO [B-12-R-24]
Describe the mechanical and chemical digestion in the oral cavity.

MECHANICAL AND CHEMICAL DIGESTION IN THE ORAL CAVITY

The process of digestion begins not in the stomach, but in the **oral cavity** — the mouth. Far from being a passive entryway, the mouth is a **highly active site** where food undergoes both **mechanical and chemical changes** that are essential for the rest of the digestive process to proceed smoothly. Without efficient digestion at this initial stage, later processes such as absorption and assimilation become less effective.

Oral Cavity

The oral cavity, also referred to as the buccal cavity, is bounded externally by the lips and cheeks, *and* internally by the tongue and palate. It contains a chamber between the palate and the tongue, where food is received and processed during the initial stages of digestion.

The **tongue** is a muscular organ whose surface is covered with projections called **papillae**. These structures increase friction, aiding in the manipulation of food, and many of them house **taste buds** responsible for detecting different taste modalities. The **palate** forms the roof of the oral cavity and separates it from the nasal passages.

The **teeth**, embedded in the upper and lower jaws, are specialized for cutting, tearing, and grinding food, facilitating its mechanical breakdown. The oral cavity also contains **three pairs of salivary glands**—the parotid, submandibular, and sublingual glands—which secrete **saliva**. Saliva lubricates the food, initiates chemical digestion, and contributes to oral hygiene.

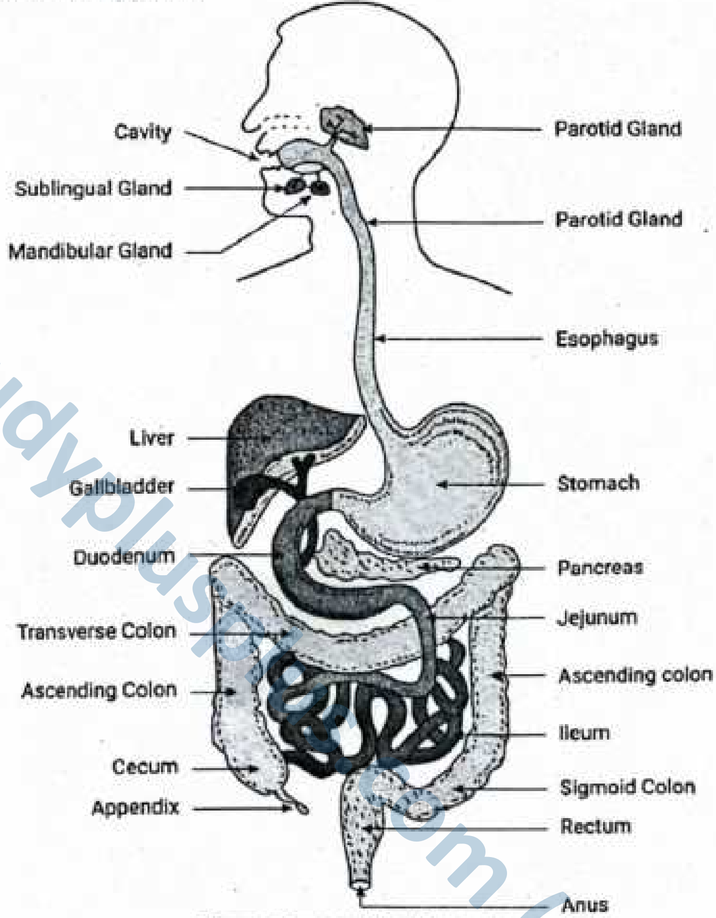

Figure: Human digestive system

Mechanical Digestion: Breaking Food Down Physically

Mechanical digestion in the oral cavity is primarily carried out by the **teeth** and assisted by the **tongue**, **cheeks**, and **palate**. This process is responsible for physically **breaking large chunks of food into smaller pieces**, increasing the total surface area for enzyme action.

▶ The Role of Teeth

Human teeth are specialized for various functions:

- ***Incisors*** (front teeth) are flat and sharp — ideal for cutting food.
- **Canines** (pointed teeth) are adapted for tearing.
- **Premolars and molars** have flat, ridged surfaces to grind and crush food thoroughly.

This coordinated activity is called **mastication**. The up-and-down and side-to-side movements of the lower jaw, combined with the force of the cheek muscles, help crush food into a semi-solid mass called a **bolus**.

> **Interesting Note**
>
> The efficiency of chewing directly affects digestion. Food that is not chewed properly may pass into the stomach in large pieces, delaying digestion and sometimes causing discomfort or indigestion.

▶ The Tongue: Not Just for Taste

The **tongue** plays a dynamic role. It constantly moves food between the teeth, gathers the chewed pieces, and mixes them with saliva. As food is chewed, the tongue shapes it into a **bolus** and pushes it to the back of the mouth for swallowing. The tongue also helps press the bolus against the hard palate during chewing.

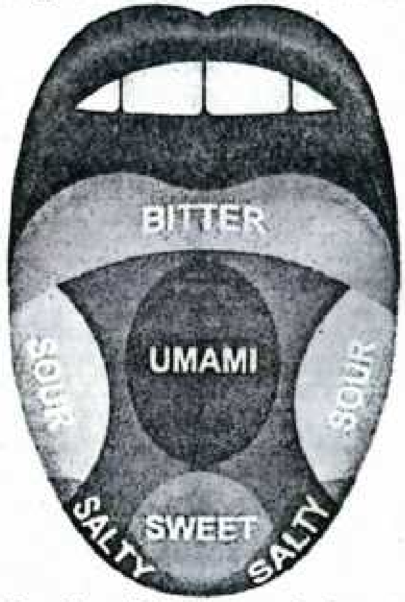

Figure: Location of Human taste buds on tongue

Chemical Digestion: Enzymes Begin Their Work

While mechanical digestion breaks food into smaller physical pieces, it is **chemical digestion** that transforms complex molecules into absorbable forms. This chemical process begins **right in the oral cavity**, even before food reaches the stomach — thanks to the **enzymatic action of saliva.**

▶ The Role of Saliva in Digestion

Saliva is more than just a watery fluid; it is a **biochemical cocktail** essential for initiating digestion, lubricating the bolus, maintaining oral hygiene, and even enabling taste. Saliva is secreted by **three pairs of major salivary glands,** each located in specific regions of the oral cavity:

- **Parotid glands:** These are the **largest salivary glands**, situated just in front of and slightly below each ear. They open into the oral cavity near the upper second molars. Their secretions are rich in **salivary amylase,** making them key players in starch digestion.
- **Submandibular glands**: Located beneath the mandible (lower jaw), these glands produce a **mixed secretion** — both serous (watery) and mucous. They contribute the **majority** of daily saliva volume (~65-70%) and are critical in moistening and lubricating food.
- **Sublingual glands:** Found beneath the tongue, these are the **smallest** salivary glands. Their secretion is primarily **mucous,** aiding in lubrication and the formation of a smooth bolus.

A healthy adult produces between **1.0 to 1.5 liters of saliva per day,** depending on factors such as hydration, diet, emotional state, and time of day. Most of this is produced **in response to food stimuli** — taste, smell, or even thought — via **reflex control from the nervous system.**

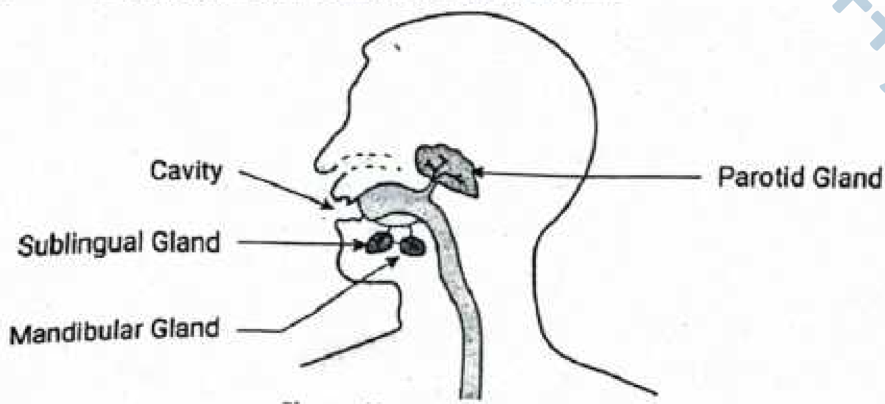

Figure: Human Salivary Glands

- **Composition of Saliva: More Than Just Water**

Though saliva is **98–99.5% water**, the remaining components are highly significant:

Component	Function
Salivary amylase	Initiates chemical digestion of starch into maltose
Lingual lipase	Begins digestion of triglycerides (activated more in stomach)
Mucus	Lubricates food, aiding bolus formation and ease of swallowing
Lysozyme	Enzyme with antibacterial properties — breaks bacterial cell walls
IgA antibodies	Provide immune defense in the oral cavity
Electrolytes (Na^+, Cl^-, HCO_3^-)	Help maintain pH (6.7–7.0), activate enzymes, and maintain ionic balance
Epidermal Growth Factor (EGF)	Promotes healing and maintenance of oral mucosa

- **Other Functions of Saliva**

In addition to digestion:

- **Lysozyme** in saliva **kills bacteria**, providing a **first line of defense** against pathogens entering with **food**.
- Saliva **cleanses the mouth** and helps with **taste perception**, as only dissolved molecules can be tasted by the tongue's receptors.
- The mucus in saliva ensures that the bolus moves smoothly down the esophagus without damaging the lining.

Enzymatic Action: Starting Carbohydrate and Fat Digestion

- **Salivary Amylase (Ptyalin)**

The most functionally significant enzyme in the oral cavity is **salivary amylase**, also known as **ptyalin**. It acts on **starch**, a polysaccharide composed of glucose units, and breaks it down into **maltose**, a disaccharide.

Reaction:

Starch → Maltose

(catalyzed by salivary amylase)

- Optimal activity: Neutral pH (6.7–7.0)
- Inactivated upon reaching the **acidic stomach** (pH ~2)

Only **partial digestion of starch** occurs in the mouth due to the brief time food stays there. Yet, this early breakdown is crucial in easing the digestive load on the small intestine.

- **Lingual Lipase**

Although present in smaller quantities in adults, **lingual lipase** is secreted **by Ebner's glands** located on the dorsal surface of the tongue. This enzyme **initiates lipid digestion**, especially triglycerides, but becomes **more active in the acidic environment** of the stomach (pH ~4–5). It plays a larger digestive role in **neonates and infants**, where milk fat forms a major dietary component.

In infants, lingual lipase is vital for milk fat digestion, compensating for their underdeveloped pancreatic enzyme production.

- **Nervous Control of Saliva Secretion**

Salivary secretion is under autonomic nervous system control, mostly parasympathetic:

- Stimuli like sight, smell, taste, and thought of food trigger reflex pathways originating in the medulla oblongata.
- Stimulation of parasympathetic fibers (cranial nerves VII and IX) increases watery saliva flow.
- Sympathetic stimulation, in contrast, produces **thicker, enzyme-poor saliva,** often seen during stress.

Stress and Dry Mouth: During anxiety or exams, sympathetic dominance may cause **dry mouth** (xerostomia), reducing enzyme effectiveness and making swallowing difficult.

DID YOU KNOW?

- Saliva **begins digestion, protects oral tissues, assists in speech**, and even helps **taste food**, as only dissolved molecules can bind to taste receptors.
- Salivary flow follows a **circadian rhythm**, being higher during meals and lower at night. This is why **morning breath** occurs — decreased saliva during sleep allows bacterial growth.
- The phrase **"mouth-watering"** isn't just poetic — it's biological. The **sight or smell of food triggers saliva production**, preparing the body for digestion.
- People with **dental problems** often suffer from digestive issues because they cannot chew food properly, emphasizing the importance of oral health in overall nutrition.

Integration of Mechanical and Chemical Processes

Both mechanical and chemical digestion in the mouth are tightly integrated. The more thoroughly the food is chewed, the greater the **surface area** exposed to enzymes, allowing for **faster and more efficient digestion**. Food that is not properly chewed may reach the stomach intact, slowing down the digestive process and increasing the burden on gastric enzymes.

Compare the Mechanical and Chemical Digestion in Oral Cavity

Feature	Mechanical Digestion	Chemical Digestion
Definition	Physical process that breaks down food into smaller pieces without altering its chemical structure.	Chemical process that breaks down complex food molecules into simpler, absorbable forms through enzymatic action.
Primary Purpose	To increase the surface area of food for enzyme action.	To convert large, insoluble molecules into smaller, soluble ones for absorption.
Main Processes Involved	Chewing (mastication), grinding, mixing by tongue and peristaltic movement.	Enzymatic hydrolysis of carbohydrates, lipids, and proteins.
Key Organs Involved	Teeth, tongue, cheeks, stomach muscles.	Salivary glands, stomach, pancreas, small intestine.
Occurrence in Oral Cavity	Carried out by teeth and tongue to crush and mix food.	Initiated by salivary amylase and lingual lipase.
End Result	Food broken into smaller physical pieces; forms a bolus.	Food molecules partially broken down into simpler compounds like maltose or lipids.

SLOs Based (MCQs)

ORAL CAVITY, MECHANICAL & CHEMICAL DIGESTION

1. Which of the following structures increases friction and houses taste buds on the tongue?
A. Villi B. Papillae
C. Alveoli D. Cilia

2. Which enzyme initiates the breakdown of starch in the oral cavity?
A. Pepsin B. Trypsin
C. Salivary amylase D. Maltase

3. The salivary glands responsible for secreting saliva include:
A. Pituitary, thyroid, adrenal
B. Parotid, submandibular, sublingual
C. Parotid, pancreatic, lingual
D. Mandibular, hepatic, lacrimal

4. Which statement about lingual lipase is correct?
A. It fully digests proteins in the mouth.
B. It hydrolyzes cellulose in the stomach.
C. It begins triglyceride digestion and is more active in the stomach.
D. It converts starch into glucose directly.

5. Mechanical digestion in the oral cavity is mainly achieved by:
A. Enzyme secretion
B. Swallowing
C. Chewing by teeth and manipulation by tongue
D. Absorption by blood vessels

Answer Key

1. B	2. C	3. B	4. C	5. C

Short Questions

Q.1 Why is an increase in surface area important during mechanical digestion in the oral cavity?

Ans. Chewing breaks food into smaller pieces, increasing its surface area. This allows digestive enzymes to access more food particles, improving the efficiency and speed of chemical digestion. It also facilitates the effective action of salivary enzymes, such as amylase, and prepares the bolus for further digestion in the stomach and intestines.

MARKING SCHEME (3 Marks)
- Link between mechanical digestion and increased surface area – 1.5 marks
- Importance of increased surface area for enzyme action – 1.5 marks

Q.2 How does the oral cavity contribute to both mechanical and chemical digestion?

Ans. The oral cavity carries out mechanical digestion through the action of the teeth, tongue, and cheeks, which grind and mix food. Chemical digestion begins here with salivary amylase breaking down starch into maltose, and lingual lipase initiating lipid digestion. These processes prepare the food for efficient digestion further along the alimentary canal.

MARKING SCHEME (3 Marks)
- Parts of oral cavity involved in mechanical digestion – 1 mark
- Identification of mechanical digestion processes – 1 mark
- Agents of chemical digestion in the oral cavity – 1 mark

Q.3 What would happen if the salivary glands failed to secrete saliva?

Ans. Without saliva, food would remain dry, making chewing and swallowing difficult. Early starch breakdown by salivary amylase would not occur, and the lack of lubrication would reduce oral hygiene, increasing the risk of tooth decay, gum disease, and other oral infections.

MARKING SCHEME (3 Marks)
- Food would remain dry – 1 mark
- Early starch breakdown prevented – 1 mark
- Reduced lubrication and compromised oral hygiene – 1 mark

Q.4 Describe the functional advantage of having both α-1,4 and α-1,6 glycosidic bond hydrolysis in starch digestion.

Ans. Salivary amylase primarily hydrolyses α-1,4 bonds to yield maltose and maltotriose, but can also act on α-1,6 linkages to form isomaltose. The ability to break both types of bonds allows for more complete starch digestion, producing simpler sugars that can be further digested and absorbed in the small intestine.

MARKING SCHEME (3 Marks)
- Naming products after hydrolysis – 2 marks
- Explanation of how dual enzyme action improves digestion and absorption – 1 mark

Q.5 Why is lingual lipase not fully effective in the oral cavity?

Ans. Lingual lipase begins triglyceride digestion but is only slightly active at the neutral pH of the oral cavity. It functions optimally in the acidic pH of the stomach, where it significantly contributes to lipid breakdown. This staged activity ensures a gradual and efficient digestion of fats.

MARKING SCHEME (3 Marks)
- pH of the oral cavity – 1 mark
- pH of the stomach – 1 mark
- Optimal pH for lingual lipase activity – 1 mark

SLO [B-12-R-25]
Explain swallowing and peristalsis.

SWALLOWING AND PERISTALSIS

Once food has been **mechanically and chemically processed in the mouth** into a soft, moist bolus, it must travel from the oral cavity to the stomach. This journey involves **two highly coordinated processes**: **swallowing (deglutition)** and **peristalsis**.

These processes are not merely passive movements — they involve **precise muscular coordination, reflex actions, and nervous system control** to ensure food reaches the stomach quickly and safely, without entering the respiratory tract.

Swallowing (Deglutition)

Swallowing is the process by which the bolus moves from the **mouth** to the **stomach**. It occurs in **three main phases**, some voluntary and others involuntary.

- **Phase 1: Oral Phase (Voluntary Phase)**

  The first stage of swallowing is the only one under conscious control. In this phase, the tongue pushes the **bolus** firmly against the **hard palate** and propels it toward the **oropharynx**. Food at this stage is moistened with saliva, shaped into a manageable mass, and positioned for swallowing. Once the bolus reaches the back of the mouth, **sensory receptors** in the **oropharynx** are stimulated, triggering the **swallowing reflex**. This reflex will carry the bolus through the next stages without further voluntary effort. The process begins with **lip closure**, preventing any loss of food from the mouth.

- **Phase 2: Pharyngeal Phase**

  This second stage is **involuntary** and remarkably rapid, lasting only about one second. As soon as the swallowing reflex is triggered, the **soft palate** elevates to close off the **nasopharynx**, preventing food or liquid from entering the nasal cavity. Simultaneously, the **larynx** moves upward while the **epiglottis** folds down to cover the **glottis**, ensuring that food is directed away from the airway. At the same time, the **vocal cords** contract to further seal the tracheal opening, temporarily halting breathing. The **upper oesophageal sphincter (UES)** then relaxes, allowing the bolus to pass from the pharynx into the oesophagus.

> **Clinical Insight**
>
> If the epiglottis does not close properly, food or liquid can enter the airway — a condition known as aspiration — which triggers coughing or choking as the body attempts to clear the obstruction.

- **Phase 3: Esophageal Phase**

  In the final stage, the bolus enters the **esophagus**, where a series of involuntary muscular contractions known as **peristaltic waves** push it toward the stomach. These coordinated contractions ensure that the bolus moves in a single direction, regardless of body position — even if one is upside down. As the bolus approaches the stomach, the **lower esophageal sphincter (LES)** relaxes at precisely the right moment, allowing smooth entry into the stomach while preventing stomach contents from flowing back into the esophagus.

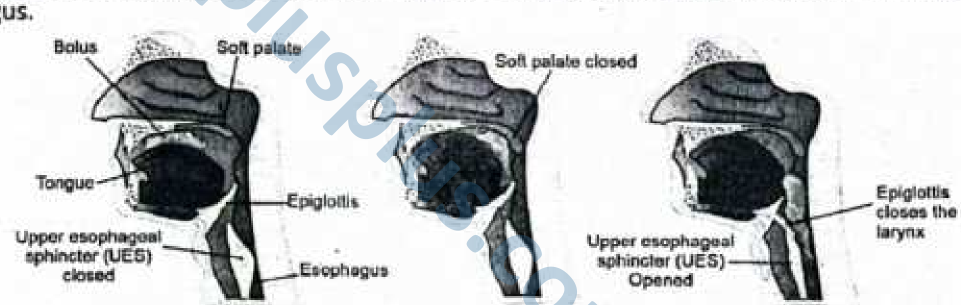

Figure: Process of Swallowing

Peristalsis: The Wave of Movement

Peristalsis is a series of coordinated, rhythmic contractions of the smooth muscle lining the esophagus and, later, other parts of the gastrointestinal tract. It is an involuntary, wave-like muscular contraction that propels contents through the digestive tract and other tubular systems. This process facilitates forward (anterograde) movement starting from the esophagus through to the intestines, ensuring efficient transport, mixing, and elimination of food and fluids.

- **Mechanism of Peristalsis**

  The process begins when circular muscles contract behind the bolus, preventing backflow, while longitudinal muscles contract ahead of the bolus, shortening the pathway and pulling the food forward. These alternating contractions create a wave-like motion that propels the bolus toward the stomach. Muscle contractions occur in a stepwise manner: a wave of smooth muscle relaxation occurs ahead of the bolus, allowing the tract to expand, followed by a wave of contraction behind the bolus, pushing it forward. This mechanism is under involuntary control, regulated by the myenteric plexus located between the circular and longitudinal muscle layers, which governs gastrointestinal motility.

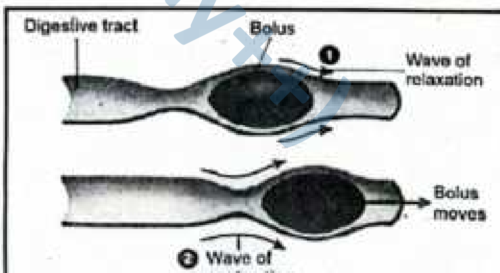

Figure: Peristalsis

Characteristics of Peristalsis

Primary peristalsis is initiated by swallowing and moves the bolus along the esophagus in about eight to ten seconds. Secondary peristalsis is triggered by irritation or if the bolus becomes lodged, continuing until the obstruction is cleared. Although gravity can assist peristalsis when the body is upright, the process remains effective even in zero gravity — a fact demonstrated during space missions. Astronauts can swallow food normally in space because peristalsis functions independently of gravity.

Functions of Peristalsis

Peristalsis serves several key roles in digestion and bodily function. In propulsion, it moves food and fluids through the digestive tract, supporting digestion and absorption. In mixing, particularly within the stomach and small intestine, it blends food with digestive secretions and increases contact with absorptive surfaces. In waste removal, peristaltic waves in the large intestine push waste toward the rectum for elimination. Beyond the digestive system, peristalsis also occurs in the urinary tract, moving urine from the kidneys to the bladder, and in the bile ducts, transporting bile to the duodenum.

Anti-Peristalsis

In contrast, anti-peristalsis refers to the reverse movement of contents in the digestive tract, typically associated with the vomiting reflex. This movement occurs from the intestine toward the stomach or mouth and is usually a defensive mechanism to expel harmful substances. While peristalsis is a normal, continuous process essential for digestion, anti-peristalsis is an abnormal or reflexive response that may result from irritation, toxins, or psychological factors. Examples include vomiting due to food poisoning or regurgitation in ruminants.

Comparison of Peristalsis and Anti-Peristalsis

Peristalsis is a coordinated, rhythmic contraction of smooth muscles moving contents in the aboral direction, from the mouth toward the anus, under the regulation of the autonomic and enteric nervous systems. It supports digestion, nutrient absorption, and waste elimination. Anti-peristalsis, on the other hand, moves contents in the oral direction, from the intestine toward the stomach or mouth, often as a protective reflex. While peristalsis promotes normal digestive function, anti-peristalsis is a reaction to potentially harmful stimuli and serves to remove them from the body.

Comparison of Peristalsis and Anti-Peristalsis

Feature	Peristalsis	Anti-Peristalsis
Direction of Movement	Aboral (from mouth toward anus)	Oral (from intestine toward stomach/mouth)
Nature	Normal, continuous process essential for digestion	Abnormal or reflexive process, usually defensive
Control	Autonomic and enteric nervous systems	Reflex pathways triggered by irritation, toxins, or stimuli
Function	Propels food, mixes with digestive secretions, aids absorption, moves waste	Expels harmful substances from stomach or intestines
Occurrence	Throughout digestive tract, urinary tract, bile ducts	Mainly in vomiting, regurgitation, or some gastrointestinal disorders
Example	Swallowing and moving chyme through intestines	Vomiting after food poisoning

Neural Control of Swallowing and Peristalsis

The process of swallowing is coordinated by a **swallowing centre** located in the **medulla oblongata** and the lower part of the **pons**. This centre receives sensory input from the oral cavity and pharynx and integrates it to produce the precise motor sequences required for the safe passage of food and drink.

Several **cranial nerves** are involved in this process:

- The Trigeminal nerve (V) assists in mastication and sensation from the face and oral cavity.
- The Facial nerve (VII) controls movements of facial expression and contributes to saliva secretion.
- The Glossopharyngeal nerve (IX) is involved in taste, sensation from the pharynx, and initiation of the *swallowing reflex.*
- The Vagus nerve (X) plays a key role in pharyngeal and laryngeal movements, as well as peristalsis in the oesophagus.
- The Hypoglossal nerve (XII) controls tongue movements necessary for bolus propulsion.

Although swallowing is primarily a reflex action, its initiation can be voluntary, allowing an individual to consciously decide when to start the process. Once initiated, the reflex ensures an efficient and safe transfer of the bolus to the stomach.

Role of Sphincters in Swallowing and Peristalsis

Two muscular rings—known as **oesophageal sphincters**—regulate the entry and exit of food along the oesophagus.

The **Upper Oesophageal Sphincter (UES)** is normally closed, preventing air from entering the oesophagus during breathing. It relaxes briefly during swallowing to allow the bolus to pass into the oesophagus.

The **Lower Oesophageal Sphincter (LES)** is located at the junction between the oesophagus and the stomach. Its primary role is to prevent the backflow of acidic gastric contents into the oesophagus. It remains closed under resting conditions and only relaxes as the bolus approaches the stomach.

Weakness of the LES can lead to **gastro-oesophageal reflux disease (GORD),** a condition where stomach acid flows back into the oesophagus, causing irritation, heartburn, and potential damage to the mucosal lining.

Swallowing and Peristalsis

Process	Location	Type of Muscle	Control	Function
Voluntary phase	Oral cavity	Skeletal muscle	Voluntary	Move bolus to pharynx
Pharyngeal phase	Pharynx	Skeletal muscle	Reflexive	Prevent food entering airway
Esophageal phase	Esophagus	Skeletal + smooth muscle	Reflexive	Propel bolus to stomach
Peristalsis	Esophagus	Smooth muscle	Reflexive	Continuous propulsion of bolus

SLOs Based (MCQs)

SWALLOWING AND PERISTALSIS

1. Which phase of swallowing is under voluntary control?
A. Pharyngeal phase B. Esophageal phase
C. Oral phase D. Gastric phase

2. What prevents food from entering the nasal cavity during swallowing?
A. Epiglottis B. Uvula
C. Soft palate D. Vocal cords

3. Which of the following best defines peristalsis?
A. Voluntary movement of skeletal muscles in digestion
B. Involuntary rhythmic contractions of smooth muscles
C. Reflexive closure of the airway during swallowing
D. Contractions that mix blood in vessels

4. Which structure coordinates peristaltic movement in the gut?
A. Hypothalamus B. Myenteric plexus
C. Medulla oblongata D. Vagus nerve

5. Which of the following events occurs during the pharyngeal phase?
A. Voluntary pushing of bolus by the tongue
B. Movement of food through the esophagus
C. Blocking of nasal cavity by the soft palate
D. Absorption of nutrients

6. Anti-peristalsis is most closely associated with which process?
A. Defecation B. Vomiting
C. Absorption D. Swallowing

7. Which two muscle layers are involved in peristaltic motion?
A. Skeletal and cardiac B. Smooth and skeletal
C. Circular and longitudinal D. Oblique and circular

Answer Key

1. C	2. C	3. B	4. B	5. C	6. B	7. C

Short Questions

Q.1 *Identify and explain the three phases of swallowing.*

Ans. Swallowing consists of three sequential phases: **oral, pharyngeal,** and **oesophageal.**

- ***Oral* phase:** Voluntary; food is chewed, mixed with saliva, and formed into a bolus. The tongue pushes the bolus to the back of the mouth while the lips close.

MARKING SCHEME (3 Marks)
- Identification of all three phases (1.0)
- *Accurate description of each phase* (1.5)
- Inclusion of voluntary/involuntary control (0.5)

- **Pharyngeal phase:** Involuntary; triggered when the bolus reaches the pharynx. The soft palate rises to block the nasal cavity, the epiglottis covers the laryngeal opening, and respiration is momentarily paused.
- **Oesophageal phase:** Involuntary; peristaltic contractions move the bolus down the oesophagus toward the stomach. The lower oesophageal sphincter relaxes to allow entry into the stomach.
- These phases ensure safe, coordinated movement of food from mouth to stomach while protecting the airway.

Q.2 Describe the process occurring during the oral phase of swallowing.

Ans. The oral phase is the **voluntary** stage of swallowing. Food is first broken down by chewing and moistened with saliva to form a soft bolus. The lips close to prevent food escape, and the tongue plays a central role by shaping and positioning the bolus. The tongue then presses the bolus against the hard palate and propels it towards the oropharynx. This movement initiates the swallowing reflex by stimulating sensory receptors in the oropharyngeal region. The oral phase ends when the bolus passes beyond the tongue's posterior margin, entering the pharyngeal phase. This phase requires precise coordination between jaw, tongue, and facial muscles to ensure effective transfer of food while avoiding premature entry into the throat.

MARKING SCHEME (3 Marks)
- Mention of voluntary control (0.5)
- Formation and positioning of bolus (1.0)
- Propulsion of bolus to oropharynx and reflex initiation (1.5)

Q.3 Explain how the airway is protected during swallowing.

Ans. Airway protection occurs mainly during the **pharyngeal phase**. As the bolus enters the pharynx, the **soft palate** elevates to close the nasopharynx, preventing food from entering the nasal cavity. Simultaneously, the **epiglottis** folds down over the laryngeal opening to shield the trachea. The **vocal cords** contract tightly, further sealing the airway and preventing aspiration. These actions are coordinated by the swallowing centre in the medulla oblongata and are accompanied by a brief cessation of breathing, known as **deglutition apnoea**. This pause ensures no inhalation occurs while the bolus passes over the airway opening. The sequence is rapid and reflexive, resuming normal breathing once the bolus has cleared the pharynx.

MARKING SCHEME (3 Marks)
- Role of soft palate (0.75)
- Role of epiglottis (0.75)
- Role of vocal cords and breathing pause (1.5)

Q.4 Explain the role of peristalsis in digestion.

Ans. Peristalsis is an **involuntary, rhythmic wave of contraction and relaxation** of smooth muscle in the walls of the digestive tract. It begins in the oesophagus after swallowing and continues through the stomach, small intestine, and large intestine. In the **oesophagus**, *peristalsis* propels the bolus toward the stomach. In the **small intestine**, it mixes chyme with digestive juices and moves it along for nutrient absorption. In the **large intestine**, peristalsis moves undigested material toward the rectum. The movement is coordinated by the enteric nervous system, particularly the myenteric plexus, ensuring a one-way flow and preventing backflow under normal conditions. Peristalsis thus plays a key role in both the propulsion and mechanical processing of food, aiding digestion and waste elimination.

MARKING SCHEME (3 Marks)
- Definition and involuntary nature (0.75)
- Role in oesophagus and small intestine (1.25)
- Role in large intestine and prevention of backflow (1.0)

Q.5 Describe the function of the myenteric plexus in peristalsis.

Ans. The **myenteric plexus** (Auerbach's plexus) is a network of neurons located between the **circular** and **longitudinal** muscle layers of the gastrointestinal tract. Its primary role is to regulate gut **motility** by coordinating smooth muscle contractions. During peristalsis, it stimulates contraction of the circular muscles behind the bolus and relaxation ahead of it, producing a wave that propels food forward. It operates mainly under reflex control but can be influenced by the autonomic nervous system. The myenteric plexus ensures the rhythmic, directional movement of food while preventing uncoordinated contractions. It is essential for efficient transport, mixing, and emptying of digestive contents throughout the tract.

MARKING SCHEME (3 Marks)
- Location between muscle layers (0.75)
- Role in coordinating contractions (1.25)
- Influence on directional and rhythmic movement (1.0)

Q.6 Define anti-peristalsis and state when it occurs.

Ans. Anti-peristalsis is the **reverse movement** of contents within the gastrointestinal tract, opposite to the normal direction of peristaltic waves. Instead of propelling material toward the anus, contractions move it backward. This mechanism is most commonly triggered during **vomiting**, when strong reverse contractions of the small intestine and stomach force contents back into the oesophagus and out through the mouth. Anti-peristalsis can also occur in certain pathological conditions or as a reflex response to toxins and irritants in the gut. While it is a protective mechanism for expelling harmful substances, frequent or severe episodes may indicate an underlying gastrointestinal disorder.

MARKING SCHEME (3 Marks)
- Definition of reverse movement (1.0)
- Main occurrence during vomiting (1.0)
- Mention of protective/pathological context (1.0)

Q.7 Explain how peristalsis contributes to waste removal.

Ans. In the **large intestine,** peristalsis plays a central role in transporting undigested material and waste products toward the rectum. After nutrients and water are absorbed, remaining material becomes more solid, forming faeces. **Slow, regular peristaltic waves** mix contents and gradually move them along the colon. **Mass peristalsis**, a stronger wave occurring a few times daily, propels faeces into the rectum, initiating the defecation reflex. This movement ensures timely waste elimination, prevents excessive bacterial overgrowth, and reduces the risk of constipation. The process is largely under involuntary control, although defecation can be voluntarily delayed by contracting the external anal sphincter.

MARKING SCHEME (3 Marks)
- Role in moving waste in large intestine (1.0)
- Mass peristalsis and defecation reflex (1.25)
- Involuntary control with voluntary delay (0.75)

Q.8 Identify the structures that move to prevent aspiration during swallowing and explain their action.

Ans. Three key structures prevent aspiration during swallowing:

1. **Soft palate** – elevates to close off the nasopharynx, stopping food from entering the nasal cavity.
2. **Epiglottis** – folds over the laryngeal inlet to block the airway opening.
3. **Vocal cords** – adduct tightly to seal the glottis.
4. These movements occur in the pharyngeal phase under reflex control from the swallowing centre in the medulla oblongata. Simultaneously, breathing pauses briefly (deglutition apnoea) to prevent inhalation of food particles. The coordinated action of these structures ensures safe passage of food into the oesophagus while protecting the respiratory tract.

MARKING SCHEME (3 Marks)
- Identification of all three structures (1.25)
- Explanation of each movement (1.25)
- Mention of breathing pause for protection (0.5)

SLO [B-12-R-26]
Illustrate with a diagram the structure of the stomach and relate each component with mechanical and chemical digestion in the stomach.

THE STRUCTURE OF THE STOMACH AND ITS ROLE IN DIGESTION

The **stomach** is a muscular, J-shaped organ situated in the upper left region of the abdominal cavity, just below the diaphragm. It acts as a temporary storage site for ingested food and plays a crucial role in both **mechanical** and **chemical** digestion before the chyme passes into the small intestine.

External Features

The **stomach** is a hollow, muscular organ situated in the upper left region of the abdominal cavity, just below the diaphragm. It serves as an expandable reservoir for ingested food and is a major site for both **mechanical** and **chemical digestion**. Anatomically, it lies between the **oesophagus** and the **duodenum** of the small intestine, and its shape can vary depending on the volume of its contents and the tone of its muscular walls.

▸ Location and Orientation

The stomach occupies the epigastric, umbilical, and left hypochondriac regions of the abdomen. Its position is obliquely placed, with its upper end (cardiac region) situated higher and to the left, and its lower end (pyloric region) angled downwards and to the right. The stomach is partly covered by the lower ribs, which help protect it from external injury.

- **Cardiac Orifice**

The **cardiac orifice** is the opening where the oesophagus joins the stomach. It is named for its proximity to the heart, not because it has any functional relation to the cardiac muscle. This opening is guarded by the **lower oesophageal sphincter (cardiac sphincter)**, a specialised ring of smooth muscle that remains closed most of the time. The sphincter opens reflexively during swallowing to allow the bolus of food to enter the stomach, and then closes to prevent acid reflux — the backflow of acidic gastric contents into the oesophagus.

> ***Clinical Note:*** Weakness or improper functioning of the cardiac sphincter can lead to **gastro-oesophageal reflux disease (GERD)**, characterised by heartburn, regurgitation, and potential damage to the oesophageal lining.

- **Pyloric Orifice**

At the lower end of the stomach lies the **pyloric orifice**, which connects the stomach to the first part of the small intestine — the **duodenum**. This opening is regulated by the **pyloric sphincter**, a thick ring of smooth muscle that controls the passage of partially digested food (**chyme**) into the small intestine. The sphincter allows chyme to enter in small amounts, ensuring proper mixing with intestinal enzymes and preventing the backflow of intestinal contents into the stomach.

The opening and closing of the pyloric sphincter are coordinated with the stomach's muscular contractions and the digestive activity in the duodenum, ensuring the digestive process proceeds efficiently.

Internal Structure

(i) Regions of the Stomach

The stomach is anatomically divided into **four distinct regions**, each with unique structural features and specialised functions that contribute to the overall process of digestion. These regions work in a coordinated manner to store food, mix it with digestive secretions, and deliver it in a controlled fashion to the small intestine.

1. **Cardia**

   The **cardia** is a narrow zone situated immediately adjacent to the **cardiac orifice**, where the oesophagus meets the stomach. This region acts as the receiving chamber for the **bolus** of food during swallowing. Its lining contains numerous **mucus-secreting glands**, which produce alkaline mucus to protect the stomach wall from the potentially corrosive effects of acidic gastric contents and to minimise the risk of **acid reflux** into the oesophagus. Although the cardia is not a major site of digestion, it plays an important role in creating a safe transition zone between the oesophagus and the acidic environment of the stomach.

2. **Fundus**

   Located superior to the cardiac orifice, the **fundus** is a dome-shaped portion of the stomach that curves upward toward the diaphragm. The fundus primarily serves as a **storage area** for ingested food that has not yet been mixed with gastric secretions. It also acts as a site for the collection of gases released during the chemical breakdown of food. While not a primary site of digestion, the fundus maintains a slightly higher pH compared to the body of the stomach, creating conditions favourable for the initial enzymatic activity of salivary amylase before gastric acid inactivates it.

3. **Body**

   The **body** (corpus) is the central and largest region of the stomach, making up the majority of its volume. It is the principal site for **mechanical digestion**, where muscular contractions churn the food, and **chemical digestion**, where gastric glands secrete **hydrochloric acid ($HC\ell$)**, **pepsinogen**, and **mucus**. $HC\ell$ lowers the pH, creating an acidic environment that activates pepsinogen into **pepsin**, an enzyme that begins the breakdown of proteins. The body's vigorous peristaltic movements ensure thorough mixing of the ingested material with gastric juice, producing a semi-liquid mixture known as **chyme**.

4. **Pylorus**

   The **pylorus** is the funnel-shaped distal region of the stomach that leads to *the **duodenum**, the first part* of the small intestine. Structurally, it is divided into the **pyloric antrum** (wider portion) and the **pyloric canal** (narrower terminal segment). The pylorus plays a crucial role in regulating the passage of chyme into the small intestine via the **pyloric sphincter**, a muscular valve that opens intermittently to release small amounts of chyme. This controlled release ensures that the duodenum receives material at a rate it can effectively process, preventing overload and allowing optimal mixing with bile and pancreatic enzymes.

Region	Description	Role in Digestion
Cardia	Narrow area near the cardiac orifice.	Receives bolus from the oesophagus; mucus glands protect the stomach lining from acidic reflux.
Fundus	Dome-shaped portion above the level of the cardiac orifice.	Stores food and releases gases produced during digestion.
Body	Central, largest portion.	Main site for mixing food with gastric juices; contains numerous gastric glands for enzyme and acid secretion.
Pylorus	Funnel-shaped distal region leading to the small intestine.	Regulates chyme release through the pyloric sphincter into the duodenum.

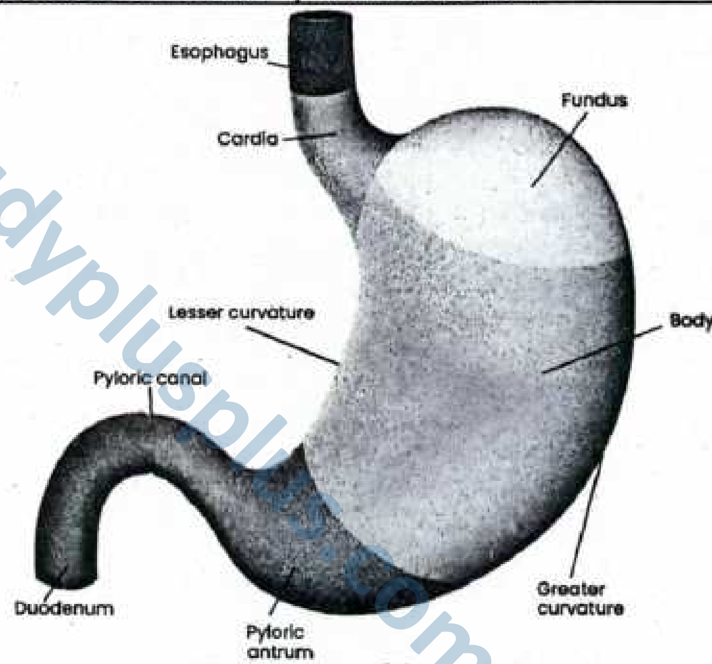

Figure: Division of the stomach

(ii) Layers of the Stomach Wall

The stomach wall consists of **four distinct layers**, arranged from the innermost surface facing the lumen to the outer protective covering. Each layer has specialised structures and functions that together enable the stomach to perform **mechanical breakdown**, **chemical digestion**, and **protection** from self-digestion.

1. **Mucosa –** ***The Functional Interface***

   The **mucosa** is the innermost layer, directly exposed to the stomach's contents. It is lined with **simple columnar epithelium** that dips down into **gastric pits**, which lead to numerous **gastric glands** embedded in the mucosa. These glands contain several types of specialised cells:
   - **Mucous cells** – Secrete thick, alkaline mucus that coats the stomach lining, forming a barrier against the corrosive effects of hydrochloric acid and digestive enzymes. This protective layer is essential in preventing **gastric ulcers**.
   - **Parietal cells** – Produce **hydrochloric acid (HCℓ)**, which lowers the stomach's pH to around 1.5–3.5, creating an optimal environment for enzyme activity and killing ingested microbes. *Parietal cells also* secrete **intrinsic factor**, a glycoprotein necessary for the absorption of ***vitamin*** B_{12} in the ileum.
   - **Chief cells** – *Secrete* **pepsinogen**, an inactive enzyme precursor that is activated to **pepsin** by the *acidic* environment. Pepsin begins the chemical breakdown of proteins into peptides.
   - **Enteroendocrine cells** – Release hormones such as **gastrin**, which stimulates gastric gland secretion and enhances muscular contractions of the stomach wall.

Beneath the epithelial layer lies the **lamina propria** (a connective tissue layer housing capillaries and immune cells) and the **muscularis mucosae**, a thin band of smooth muscle that helps expel glandular secretions into the lumen.

2. **Submucosa** – ***The Support Network***

The **submucosa** is a thick layer of **areolar connective tissue** lying beneath the mucosa. It contains:

- **Blood vessels** – Supply oxygen and nutrients to the stomach wall and carry away absorbed substances.
- **Lymphatic vessels** – Play a role in immune defence and lipid transport.
- **Nerve plexuses** – The **submucosal plexus** (Meissner's plexus) regulates glandular secretions and local blood flow.

This layer forms the structural and vascular backbone of the mucosa, allowing it to function efficiently.

3. **Muscularis Externa – The Churning Engine**

The **muscularis externa** of the stomach is unique in having **three** layers of smooth muscle instead of the typical two found in most digestive organs:

(i) Outer longitudinal layer – Shortens the stomach lengthwise.
(ii) Middle circular layer – Constricts the lumen, aiding in mixing and mechanical **breakdown**.
(iii) Inner oblique layer – Unique to the stomach; allows a twisting motion that intensifies churning.

These layers work together to perform **peristalsis** and **mixing waves**, ensuring food is thoroughly broken down into chyme before passing into the duodenum.

4. **Serosa** – ***The Protective Coat***

The outermost layer, the **serosa**, is a thin membrane composed of **simple squamous epithelium** (mesothelium) over a small amount of connective tissue. It is continuous with the **visceral peritoneum**, allowing the stomach to move freely within the abdominal cavity while remaining protected from friction.

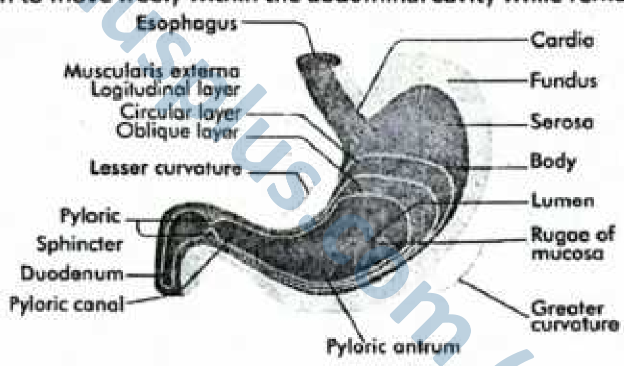

Figure: Cutaway section of the stomach reveals muscular layers and internal anatomy

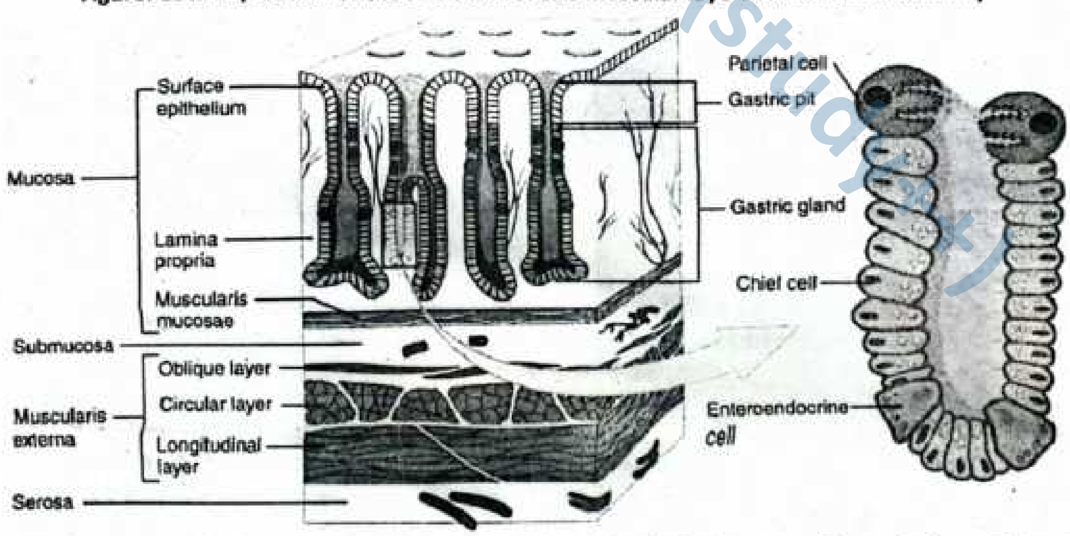

Figure: A section of stomach wall that illustrates its histology, including several gastric pits and glands

Gastric Gland Cell Types and Their Secretions

The **gastric glands** of the stomach are specialized epithelial structures located in the mucosa. They are essential for producing the secretions required for both **mechanical and chemical digestion**, as well as for protecting the stomach lining. The glands contain **four main functional cell types**, each with distinct products and roles:

1. **Surface Mucous Cells – *Protective Layer Formation***
   - **Location:** Line the **gastric pits** and cover the stomach lumen.
   - **Secretion:** Thick **alkaline mucus** rich in bicarbonate ions.
   - **Function:** Forms a protective barrier between the mucosa and the acidic gastric environment, preventing **self-digestion** by HCl and enzymes.
   - Damage to this layer can result in **gastric ulcers**.
2. **Parietal Cells (Oxyntic Cells) – *Acid and Vitamin Absorption Support***
   - **Location:** Deep in the gastric glands, especially abundant in the **fundus and body** regions.
   - **Secretions:**
     - **Hydrochloric Acid (HCℓ):** Maintains a **low pH (~1.5–3.5)**, which:
       - Kills most ingested microbes.
       - Denatures proteins, making them more accessible to enzymes.
       - Stops salivary amylase activity (shifting carbohydrate digestion to the intestine).
       - Activates pepsinogen into pepsin.
     - **Intrinsic Factor:** A **glycoprotein** essential for vitamin B_{12} absorption in the **ileum.**
   - Loss of intrinsic factor production leads to **pernicious anemia.**
3. **Chief Cells (Zymogenic Cells) – Enzyme Precursors for Protein Digestion**
   - **Location:** Primarily in the **base** of gastric glands.
   - **Secretion:**
     - **Pepsinogen:** An inactive precursor (**zymogen**) converted into **pepsin** by HCℓ.
     - **Pepsin:** An active protease that breaks peptide bonds, producing smaller polypeptides.
   - **Protective Mechanism:** Storing pepsin as an inactive form prevents **autodigestion** of the cell.
4. **Endocrine Cells (G Cells) – *Hormonal Regulation***
   - **Location:** Mainly in the pyloric antrum.
   - **Secretion:**
     - **Gastrin** – released into the bloodstream (not into the lumen).
   - **Functions of Gastrin:**
     - Stimulates **parietal cells** to secrete more HCℓ.
     - Enhances **gastric motility** for better mixing and emptying.
     - Increases activity of **chief cells.**

► **Gastric Gland Cell Types**

Cell Type	Secretion	Function
Surface Mucous	Alkaline mucus	Protects mucosa from acid/enzymes
Parietal (Oxyntic)	HCℓ, Intrinsic factor	Acid for digestion; Vitamin B_{12} absorption
Chief (Zymogenic)	Pepsinogen	Protein digestion (after activation to pepsin)
Endocrine (G)	Gastrin (hormone)	Stimulates acid secretion and motility

Mechanical Digestion in the Stomach

The stomach is uniquely adapted for **vigorous physical processing of food**. Its wall contains three layers of smooth muscle—**longitudinal, circular,** and an additional **oblique layer** found only in the stomach. The oblique layer provides extra strength for **churning movements** that cannot be achieved by the other two layers alone.

When a bolus of food enters the stomach through the cardiac orifice, **peristaltic waves** and **rotational churning** knead the food, thoroughly mixing it with **gastric juice**. This process reduces large food particles into much smaller fragments and evenly exposes them to digestive enzymes and acid.

The end product of mechanical digestion is **chyme**—a thick, semi-liquid mixture of partially digested food, water, hydrochloric acid, and enzymes. Chyme is gradually released into the **duodenum** through the **pyloric**

sphincter in small, controlled amounts, allowing the small intestine to efficiently complete digestion and begin nutrient absorption.

Chemical Digestion in the Stomach

Chemical digestion in the stomach is driven by **gastric secretions** produced by specialized cells located in the **gastric glands** of the mucosa. These secretions contain acids, enzymes, mucus, and hormones, each with specific roles in breaking down food and regulating stomach activity.

Component	Secreted By	Description	Key Functions
Hydrochloric Acid (HCℓ)	**Parietal cells**	Strong acid, pH 1.5–3.5	- Denatures proteins, making them easier for enzymes to digest. - Activates **pepsinogen** into **pepsin**. - Kills most microbes present in food. - Creates optimal acidic conditions for enzyme activity.
Pepsin	**Chief cells** (as pepsinogen)	Proteolytic enzyme	- Breaks down large protein molecules into smaller polypeptides. - Secreted as inactive pepsinogen to prevent damage to stomach cells.
Mucus	**Mucous cells**	Thick alkaline secretion	- Coats and protects the stomach lining from corrosive acid and digestive enzymes. - Maintains mucosal integrity to prevent ulcers.
Intrinsic Factor	**Parietal cells**	Glycoprotein	- Essential for absorption of vitamin B_{12} in the ileum. - Lack of intrinsic factor leads to pernicious *anaemia*.
Gastrin	**G cells** (enteroendocrine cells)	Peptide hormone	- Stimulates parietal cells to secrete more HCℓ. - Enhances gastric motility and enzyme secretion. - Promotes coordination of digestive activities.

Integration of Mechanical and Chemical Digestion

Mechanical and chemical digestion work **simultaneously** in the stomach. Churning ensures every particle of food is exposed to gastric juice, while enzymes and acid chemically break down proteins into smaller units. By the time chyme leaves the stomach, carbohydrates have undergone limited breakdown (begun in the mouth), fats remain mostly intact except for some emulsification, and proteins have been partially digested into peptides—ready for the more intensive chemical processes of the small intestine.

Coordination of Digestion

The stomach does not work continuously at full capacity—it operates in **three well-regulated phases** that coordinate its activity with other parts of the digestive system. This regulation ensures that food is processed efficiently and that the small intestine is not overloaded.

1. **Cephalic Phase – *Preparation for Digestion***
    - **Trigger:** This phase begins before food enters the stomach, stimulated by the sight, smell, taste, or even thought of food.
    - **Control:** Regulated primarily by the vagus nerve (parasympathetic nervous system).
    - **Response:** Nerve impulses stimulate gastric glands to secrete HCl, pepsinogen, and mucus in anticipation of food arrival.
    - **Example:** Smelling freshly baked bread may increase stomach activity, even if you haven't eaten yet.
2. **Gastric Phase – *Active Digestion in the Stomach***
    - **Trigger:** Begins when food **actually enters the stomach**, causing it to stretch (**distension**) and activating **chemoreceptors** that detect changes in pH.
    - **Control:** Both neural and **hormonal** mechanisms are involved.
    - ***Response:***
        - Stretch receptors stimulate stronger **peristaltic contractions.**
        - Gastrin, released by **G cells**, increases **HCℓ secretion** and enzyme production.
        - pH drops to **1.5–3.5**, optimal for pepsin activity.
    - **Note:** Proteins in food act as powerful stimulants for gastrin release.

3. Intestinal Phase – *Controlled Gastric Emptying*

- **Trigger:** Begins when **chyme enters the duodenum** through the pyloric sphincter.
- **Control:** Primarily hormonal—**secretin** and **cholecystokinin (CCK)** from the duodenum slow gastric activity.

2 - **Response:**
    - **Gastric emptying slows** to give the small intestine time to neutralize acid and fully digest nutrients.
    - **Inhibitory reflexes** prevent excessive chyme from entering the duodenum at once.
- **Reason:** This prevents the small intestine from being overwhelmed and allows optimal digestion and absorption.

▶ **Phases of Gastric Activity**

Phase	Trigger	Control Mechanism	Main Response
Cephalic	Thought, sight, smell, taste of food	Neural (vagus nerve)	Prepares stomach: ↑ gastric juice, ↑ motility
Gastric	Food in stomach (stretch + proteins)	Neural + Hormonal (gastrin)	↑ HCl, ↑ enzymes, strong churning
Intestinal	Chyme in duodenum	Hormonal (secretin, CCK) + reflexes	Slows gastric emptying, protects intestine

SLOs Based (MCQs)

STOMACH

1. Where is the esophagus located?
A. Behind the vertebral column and in front of the trachea
B. In front of the vertebral column and behind the trachea
C. Within the pleural cavity
D. In the abdominal cavity

2. Which sphincter prevents gastric reflux into the esophagus?
A. Pyloric sphincter
B. Upper esophageal sphincter
C. Lower esophageal sphincter
D. Cardiac valve

3. Which stomach region connects to the duodenum?
A. Cardia
B. Fundus
C. Body
D. Pylorus

4. Which of the following correctly lists the stomach wall layers from outer to inner?
A. Serosa → Submucosa → Mucosa → Muscularis
B. Muscularis → Serosa → Mucosa → Submucosa
C. Serosa → Muscularis → Submucosa → Mucosa
D. Mucosa → Muscularis → Submucosa → Serosa

5. Which gastric gland cell secretes hydrochloric acid?
A. Chief cell
B. Mucous cell
C. Parietal cell
D. Endocrine cell

6. What is the function of intrinsic factor secreted by parietal cells?
A. Protein digestion
B. Binds vitamin B_{12} for absorption
C. Stimulates mucus production
D. Activates salivary amylase

7. Which statement about pepsin is correct?
A. It is secreted as an active enzyme
B. It digests carbohydrates in the stomach
C. It functions best at pH 7
D. It is activated from pepsinogen by hydrochloric acid

8. What enhances mechanical digestion in the stomach?
A. Presence of villi
B. Circular folds in the stomach
C. Oblique muscle layer in muscularis externa
D. Increased submucosal thickness

Answer Key

1. B	2. C	3. D	4. C	5. C	6. B	7. D	8. C

Short Questions

Q.1 Locate and describe the structure and function of the oesophagus.

Ans. The **oesophagus** is a **25 cm-long muscular tube** situated in the **mediastinum**, **positioned behind the trachea**

and **in front of the vertebral column**. It begins at the pharynx, descends through the thoracic cavity, and passes through the diaphragm at the oesophageal hiatus before joining the stomach at the **lower oesophageal sphincter** (LES). Its primary function is to **transport food and liquids** from the pharynx to the stomach via **peristaltic contractions** of its smooth muscle walls. The LES regulates entry into the stomach and prevents backflow of acidic gastric contents. The oesophagus is lined with stratified squamous epithelium to protect against mechanical abrasion during swallowing. This structure ensures efficient, one-way food movement from mouth to stomach. 2/2

MARKING SCHEME (3 Marks)
- Accurate location and anatomical pathway (1.0)
- Transport function with peristalsis (1.5)
- LES connection and reflux prevention (0.5)

Q.2 Explain the roles of the upper and lower oesophageal sphincters in digestion.

Ans. The **upper oesophageal sphincter (UES)** is located at the junction of the pharynx and oesophagus. It remains closed at rest to prevent air from entering the digestive tract. During swallowing, it relaxes in response to food or liquid, allowing entry into the oesophagus. The **lower oesophageal sphincter (LES)** is found at the junction between the oesophagus and stomach. It relaxes to permit bolus entry into the stomach and then contracts to prevent the **backflow (reflux)** of acidic gastric contents. Together, these sphincters ensure a **one-way movement** of ingested material and protect the airway and oesophageal lining. Their coordinated action is essential for maintaining digestive efficiency and preventing conditions such as gastroesophageal reflux disease (GERD).

MARKING SCHEME (3 Marks)
- Role of UES in food entry and air prevention (1.5)
- Role of LES in entry to stomach and reflux prevention (1.5)

Q.3 Identify the anatomical regions of the stomach and state their locations.

Ans. The stomach is divided into four main regions:

1. **Cardia** – surrounds the opening where the oesophagus meets the stomach, receiving ingested material.
2. **Fundus** – a dome-shaped portion above and to the left of the cardia; often filled with swallowed air or gas.
3. **Body** – the central, largest region where most mixing and digestion occur.
4. **Pylorus** – the distal region connecting to the duodenum; subdivided into the pyloric antrum, pyloric canal, and pyloric sphincter, which regulates chyme passage.
5. These regions collectively allow storage, mechanical mixing, and controlled release of gastric contents into the small intestine.

MARKING SCHEME (3 Marks)
- Cardia location and function (1.0)
- Pylorus location and control role (1.0)
- Fundus and body location (0.5 each)

Q.4 Describe the structural composition of the stomach wall.

Ans. The stomach wall has four layers:

1. **Serosa** – the outermost protective layer of visceral peritoneum.
2. **Muscularis externa** – three layers of smooth muscle: longitudinal (outer), circular (middle), and oblique (innermost). The extra oblique layer aids in churning food.
3. **Submucosa** – connective tissue containing blood vessels, lymphatics, and nerves.
4. **Mucosa** – the innermost layer lined with simple columnar epithelium, forming gastric pits that open into gastric glands. These glands secrete mucus, enzymes, and hydrochloric acid.
5. This layered structure supports both mechanical mixing and chemical digestion, while protecting the stomach from self-digestion.

MARKING SCHEME (3 Marks)
- Serosa (0.5)
- Muscularis externa with three layers (1.0)
- Submucosa (0.5)
- Mucosa with epithelium/gastric pits (1.0)

Q.5 Identify and describe the main types of gastric gland cells and their secretions.

Ans. Gastric glands contain four primary cell types:

1. **Mucous cells** – secrete alkaline mucus to protect the mucosal lining from acidic gastric juice.
2. **Parietal cells** – release hydrochloric acid ($HC\ell$) to lower stomach pH and *intrinsic factor* for vitamin B_{12} absorption.
3. **Chief cells** – produce pepsinogen, an inactive enzyme precursor that is converted to pepsin for protein digestion.

MARKING SCHEME (3 Marks)
- Identification of mucous, parietal, chief, and G cells (1.0)
- Correct matching of each secretion (1.5)
- Stated role in digestion/protection (0.5)

4. **G cells** – endocrine cells secreting gastrin, a hormone that stimulates acid secretion and enhances gastric motility.
5. Each cell type plays a specialised role in digestion and mucosal protection, ensuring optimal enzyme activity while safeguarding stomach tissues.

Q.6 Explain the process of mechanical digestion in the stomach.

Ans. Mechanical digestion in the stomach is carried out by **rhythmic muscle contractions** called **mixing waves**, generated by the muscularis externa. These contractions occur every 20 seconds and involve longitudinal, circular, and especially the extra oblique muscle layer, which provides additional churning strength. The waves mix ingested food with gastric secretions, breaking it into smaller particles and producing a semi-fluid mixture called **chyme**. This process not only reduces particle size but also ensures thorough mixing with enzymes and acid for chemical digestion. The pyloric sphincter then regulates chyme release into the duodenum for further digestion and absorption.

> **MARKING SCHEME (3 Marks)**
> - Definition of mechanical digestion / mixing waves (1.0)
> - Role of muscle layers, including oblique (1.0)
> - Formation of chyme (1.0)

Q.7 Describe the role of hydrochloric acid in stomach function.

Ans. Hydrochloric acid ($HC\ell$), secreted by parietal cells, lowers the stomach pH to about **2.0**. This acidic environment:

1. **Denatures proteins**, unravelling their structure for enzymatic breakdown.
2. **Kills most ingested microorganisms**, acting as a barrier to infection.
3. **Inactivates salivary amylase**, stopping carbohydrate digestion begun in the mouth.
4. **Converts pepsinogen into pepsin**, the active enzyme that begins protein digestion.
5. By performing these roles, $HC\ell$ supports both digestive efficiency and protection from pathogens.

> **MARKING SCHEME (3 Marks)**
> - Contribution to low pH (1.0)
> - Protein denaturation + amylase inactivation (1.0)
> - Activation of pepsinogen (1.0)

Q.8 Explain how pepsinogen is activated and state its function.

Ans. Pepsinogen is an **inactive zymogen** secreted by **chief cells** and stored in intracellular granules. Once released into the acidic gastric lumen, it is converted into active **pepsin** by the action of hydrochloric acid. Additionally, existing pepsin molecules can autocatalytically activate more pepsinogen. Pepsin functions as a **proteolytic enzyme**, breaking down proteins into smaller peptide chains for further digestion in the small intestine. Activation only in the acidic stomach prevents unwanted protein digestion within the cells that produce it.

> **MARKING SCHEME (3 Marks)**
> - Chief cell production/secretion (1.5)
> - $HC\ell$ role in activation (1.5)

Q.9 Explain how gastric secretion is regulated in the human body.

Ans. Gastric secretion is controlled by **neural** and **hormonal** mechanisms. Sensory stimuli such as sight, smell, and taste of food trigger the medulla oblongata, sending impulses via the vagus nerve to release **acetylcholine (ACh)**. ACh directly stimulates **parietal and chief cells** and promotes gastrin release from G cells in the pyloric antrum. **Gastrin** enters the bloodstream and acts on parietal cells to further increase $HC\ell$ secretion and enhance gastric motility. This coordinated response ensures that gastric juice is secreted in anticipation of, and during, food arrival, optimising digestion.

> **MARKING SCHEME (3 Marks)**
> - Role of sensory stimuli (1.0)
> - Role of acetylcholine (1.0)
> - Role of gastrin (1.0)

SLO [B-12-R-27]
Identify the role of the nervous system and gastrin hormone on the secretion of gastric juice.

ROLE OF THE NERVOUS SYSTEM AND GASTRIN HORMONE IN GASTRIC JUICE SECRETION

Gastric juice secretion is a **highly coordinated process** involving both **neural control** and **hormonal regulation**. This ensures that *the stomach* produces the right quantity and composition of gastric juice in response to the body's digestive needs.

1. **Nervous System Regulation**

The **autonomic nervous system (ANS)**, and especially its **parasympathetic division**, exerts a dominant influence over the initiation and regulation of gastric secretions. This control is primarily mediated by the **vagus nerve (cranial nerve X)**, which activates the secretory cells of the stomach both before food enters the digestive tract and while it is present in the stomach. Neural control occurs in two major functional phases: the **cephalic phase** and the **gastric phase**.

Cephalic Phase – Initiation by the Brain

The cephalic phase represents the earliest stage of gastric regulation, beginning even before food enters the mouth. It is triggered by sensory and psychological cues such as the **sight, smell, taste, or even the mere thought of food**. These stimuli activate the hypothalamus and higher brain centers, which relay impulses to the medulla oblongata. From here, **parasympathetic fibers of the vagus nerve** carry signals to the stomach. The vagal impulses stimulate three major types of secretory cells within the gastric mucosa:

- **Parietal cells**, which secrete **hydrochloric acid (HCl)** to create the acidic environment necessary for enzymatic digestion.
- **Chief cells**, which release **pepsinogen**, the inactive precursor of the protein-digesting enzyme pepsin.
- **Surface mucous cells**, which secrete a protective alkaline mucus that coats the stomach lining, safeguarding it from self-digestion by acid and enzymes.

This anticipatory mechanism ensures that the stomach is fully primed to receive and process food as soon as it arrives, increasing digestive efficiency.

Gastric Phase – Enhancement by Local Reflexes

The gastric phase begins when food physically enters the stomach. The arrival of food causes **distension of the stomach wall**, which activates **mechanoreceptors** (stretch receptors), and the breakdown of proteins releases peptides and amino acids, which are detected by **chemoreceptors** in the mucosa.

These receptors send sensory information through the vagus nerve to the medulla, resulting in **continued parasympathetic stimulation of gastric glands**. At the same time, local neural circuits within the **enteric nervous system** are activated. These short reflexes operate entirely within the walls of the stomach and do not require input from the brain, yet they effectively reinforce and sustain secretory activity.

The combined effect of long reflexes (via the vagus nerve) and short reflexes (via the enteric plexuses) ensures that the gastric glands continue to produce an optimal mix of acid, enzymes, and mucus during active digestion. This **dual-level control** maintains a high level of secretion as long as food remains in the stomach, facilitating mechanical breakdown and chemical digestion before chyme enters the small intestine.

2. **Hormonal Regulation via Gastrin**

The secretion of gastric juice is regulated by a combination of **neural control** through the autonomic nervous system and **hormonal control** via gastrin. This dual mechanism ensures that gastric glands respond appropriately to both the anticipation and presence of food, maintaining efficient digestion while protecting the stomach lining.

Neural Regulation – Autonomic Nervous System

The **parasympathetic division** of the autonomic nervous system, acting primarily through the **vagus nerve (cranial nerve X)**, is central to initiating and sustaining gastric secretions. Neural control occurs in two main phases:

(i) **Cephalic Phase – Initiation by the Brain**

This phase begins before food enters the stomach and is triggered by the sight, smell, taste, or thought of food. Higher brain centers stimulate the medulla oblongata, which then activates vagal fibers. These impulses stimulate:

- **Parietal cells** to release hydrochloric acid (HCℓ).
- **Chief cells** to release pepsinogen.
- **Surface mucous cells** to secrete protective alkaline mucus.

This anticipatory response primes the stomach for digestion.

(ii) **Gastric Phase – Enhancement by Local Reflexes**

When food enters the stomach, **mechanoreceptors detect** stretching of the walls, and **chemoreceptors** sense peptides and amino acids from protein digestion. These signals travel via the vagus nerve and through the **enteric nervous system** (local short reflexes), increasing the activity of gastric glands. The combined reflex pathways maintain a high rate of secretion while food remains in the stomach.

Hormonal Regulation – Gastrin

The principal hormone regulating gastric juice secretion is **gastrin**, produced by **G cells** located in the pyloric antrum. Gastrin acts as a powerful stimulator of acid and enzyme secretion, linking the chemical composition of the stomach contents to the digestive output of the gastric glands.

Stimuli for Gastrin Release

Gastrin secretion is triggered by:

- The presence of **partially digested proteins** (peptides) in the stomach.
- **Stomach distension** detected by stretch receptors.
- **Vagal stimulation**, which provides a neurohormonal link between nervous and endocrine control.

Actions of Gastrin

Once released into the bloodstream, gastrin:

- Stimulates **parietal cells** to increase HCl secretion, enhancing protein denaturation and enzyme activation.
- Promotes **chief cell** secretion of pepsinogen for protein digestion.
- Enhances **gastric motility**, improving mechanical mixing and food breakdown.
- Stimulates the **growth of gastric mucosa**, preserving the stomach's long-term secretory capacity.

Integration of Neural and Hormonal Control

The nervous system and gastrin hormone act synergistically. Neural signals can directly stimulate gastric glands and also promote gastrin release via vagal stimulation. Gastrin, in turn, amplifies the effects of neural inputs by further stimulating acid and enzyme secretion. This **coordinated control** ensures that gastric juice production matches the digestive demands of the meal, optimizing digestion while preventing unnecessary secretion in the absence of food.

▶ Neural vs. Hormonal Regulation of Gastric Juice Secretion

Aspect	Neural Control (Autonomic Nervous System)	Hormonal Control (Gastrin)
Source	Parasympathetic division via vagus nerve (cranial nerve X)	**G cells** in pyloric antrum of stomach
Main Stimuli	Cephalic phase: sight, smell, taste, thought of food Gastric phase: stomach distension, peptides, amino acids	Presence of peptides in stomach, distension, vagal stimulation
Mechanism	Vagal impulses → stimulate gastric glands directly + trigger local enteric reflexes	Gastrin released into blood → binds to receptors on gastric glands
Target Cells	Parietal cells, chief cells, surface mucous cells	Parietal cells, chief cells, gastric smooth muscle, mucosal cells
Primary Effects	↑ HCL secretion ↑ pepsinogen secretion ↑ mucus secretion	↑ HCL secretion ↑ pepsinogen secretion ↑ gastric motility Stimulates mucosal growth
Phases of Action	Cephalic & gastric phases (immediate response)	Primarily gastric phase (sustained response)
Integration	Prepares and initiates secretion; can stimulate gastrin release	Enhances and prolongs secretion initiated by neural signals

3. Interaction between Nervous and Hormonal Control

The regulation of gastric juice secretion is not the result of isolated neural or hormonal actions, but rather a finely tuned **interaction between both systems** to ensure optimal digestion. This collaboration allows the stomach to respond quickly to the presence of food while sustaining digestive activity for as long as necessary.

The **vagus nerve** plays a crucial dual role. It not only directly stimulates the gastric glands—including parietal cells, chief cells, and mucous cells—to secrete their respective substances but also acts indirectly by promoting the release of the hormone **gastrin** from G cells in the pyloric region. This neurohormonal link means that the nervous system can trigger both immediate and sustained digestive responses.

Once released, **gastrin amplifies the effects of vagal stimulation** by increasing the sensitivity of gastric gland cells to neural signals. This means that parietal and chief cells respond more vigorously to vagal impulses, producing higher levels of hydrochloric acid and pepsinogen. Gastrin also stimulates gastric motility and supports the growth of the gastric mucosa, ensuring that the stomach maintains its digestive efficiency over time.

In summary, the nervous and hormonal controls of gastric secretion work in harmony, with the vagus nerve initiating and sustaining activity both directly and through gastrin release, while gastrin magnifies and prolongs the secretory response. This synergy maximises the digestive capabilities of the stomach, enabling efficient processing of ingested food.

► **Regulation of Gastric Juice Secretion**

Regulator	Source	Trigger	Effect on Stomach
Vagus nerve (parasympathetic)	Medulla oblongata	Sight, smell, taste, distension	Stimulates gastric glands; promotes gastrin release
Enteric nervous system reflexes	Local plexuses	Stretch & chemical detection	Maintain secretion during gastric phase
Gastrin hormone	G cells (pyloric antrum)	Peptides, amino acids, distension, vagal stimulation	Increases HCℓ & pepsinogen secretion; enhances motility

SLO [B-12-R-28]
Describe major actions carried out on food in the three regions of small intestine.

MAJOR ACTIONS ON FOOD IN THE THREE REGIONS OF SMALL INTESTINE

The **small intestine** is the principal site for digestion and absorption of nutrients. It is divided into three regions—**duodenum, jejunum, and ileum**—each performing specialised but interconnected roles. Together, they ensure that food is broken down into absorbable units and that nutrients are efficiently transported into the bloodstream or lymph.

1. Duodenum – Primary Site of Chemical Digestion

The **duodenum** is the **shortest segment** of the small intestine, measuring approximately **25 cm in length**, and forms a C-shaped loop immediately after the stomach's pyloric sphincter. Despite its short length, it plays a **critical role in digestion**, acting as a **central mixing chamber** where the acidic chyme from the stomach is blended with **bile, pancreatic secretions**, and **intestinal juice**. These combined secretions initiate the **final and most intensive phase of chemical digestion**.

Structural Adaptations

- **Thick mucosal lining** with folds (plicae circulares) to slow chyme movement, allowing more time for digestion.
- **Brunner's glands** in the submucosa secrete alkaline mucus to protect the intestinal lining and neutralise stomach acid.
- Close proximity to ducts from the **pancreas** and **gallbladder**, ensuring immediate mixing of digestive fluids.

Major Functions

1. **Mixing of Chyme and Digestive Secretions**
   - The duodenum receives **chyme** from the stomach in small amounts through the **pyloric sphincter.**
   - Rhythmic segmentation movements mix the chyme with secretions for maximum enzyme contact.
2. **Fat Emulsification by Bile**
   - **Bile** enters via the **common bile duct** from the **liver and gallbladder**.
   - Bile salts break down large fat globules into **tiny droplets** (emulsification), greatly increasing the surface area for enzyme action.

3. **Enzymatic Breakdown by Pancreatic Juice**
   - The **pancreatic duct** delivers an enzyme-rich fluid containing:
     - **Amylase** → Converts starch into maltose.
     - **Lipase** → Breaks down emulsified fats into fatty acids and glycerol.
     - **Proteases** (trypsin, chymotrypsin) → Split polypeptides into shorter peptide chains.
   - The pancreas also produces **nucleases** for the digestion of nucleic acids.
4. **Neutralisation of Acidic Chyme**
   - Pancreatic juice contains **bicarbonate ions that** neutralise stomach acid.
   - **Brunner's glands** add further alkaline mucus, creating an optimal pH (around 7–8) for enzyme activity.
5. **Initial Absorption**
   - Although most nutrient absorption occurs later in the small intestine, the duodenum **begins absorption** of:
     - Simple sugars (e.g., glucose)
     - Amino acids
     - Small fatty acids

- **Length & Position:** Shortest section (~25 cm), located immediately after the stomach.
- **Function:** Acts as the main mixing chamber where chyme from the stomach is combined with secretions from the pancreas, liver, and intestinal glands.
- **Key Actions:**
  - Receives **bile** from the liver and gallbladder, which emulsifies fats into small droplets for enzyme action.
  - Receives **pancreatic juice,** rich in digestive enzymes:
    - Amylase → breaks down starch into maltose.
    - Lipase → digests emulsified fats into fatty acids and glycerol.
    - Proteases (trypsin, chymotrypsin) → convert polypeptides into smaller peptides.
  - Neutralises acidic chyme using bicarbonate ions from pancreatic juice and Brunner's glands.
  - Initiates absorption of simple sugars, amino acids, and small fatty acids.

2. Jejunum – Main Site of Nutrient Absorption

The **jejunum** forms the middle portion of the small intestine, measuring about **2.5 metres** in length. It lies between the **duodenum** and the **ileum** and is suspended within the abdominal cavity by the **mesentery,** which contains blood vessels, lymphatics, and nerves supplying this region.
It is the **primary site of nutrient absorption**, with a mucosal lining adapted for maximum efficiency.

Structural Adaptations

- **Thick, highly vascular walls** to facilitate rapid transport of absorbed nutrients into the bloodstream.
- **Numerous villi and dense microvilli** (forming the brush border), creating an enormous surface area for absorption.
- Rich network of **capillaries** for absorption of sugars, amino acids, and water-soluble vitamins.
- Presence of **lacteals** (lymphatic vessels) within villi for lipid absorption.

Major Functions

1. **Completion of Chemical Digestion**
   - Brush-border enzymes embedded in the microvilli membranes carry out the **final hydrolysis of food molecules:**
     - **Maltase, lactase, sucrase** → Break down disaccharides into monosaccharides (e.g., glucose, galactose, fructose).
     - **Peptidases** → Split short peptides into individual amino acids.

2. **Rapid Nutrient Absorption**
   - Monosaccharides and amino acids are absorbed into **capillaries** via active and facilitated transport.
   - Fatty acids and monoglycerides are reassembled into triglycerides within intestinal cells, packaged into **chylomicrons**, and transported into **lacteals**.
   - Vitamins and minerals such as **calcium** and **iron** are actively absorbed, supported by the alkaline pH.

For Your Information

Mucus is secreted in large amount by duodenal glands, intestinal glands, and goblet cells. The mucus provides the wall of intestine with protection against the irritating effects of acidic chyme and against the digestive enzymes that enter the duodenum from the pancreas.

3. **Transport of Chyme**
   - Peristaltic contractions propel the chyme forward while mixing it with intestinal secretions, ensuring **continuous contact with absorptive surfaces.**

3. Ileum – Final Absorptive and Defensive Section

- **Length & Position:** Longest section (~3.5 m), ending at the ileocecal valve.
- **Function:** Specialised for absorbing nutrients not taken up in the jejunum, as well as bile salts and vitamin B_{12}.
- **Key Actions:**
  - Absorbs bile salts for recycling back to the liver (enterohepatic circulation).
  - Absorbs vitamin B_{12} bound to intrinsic factor.
  - Absorbs remaining nutrients and water.
  - Contains **Peyer's patches** (lymphoid tissue) that provide immune protection *against pathogens* in the gut.

► **Functions of Small Intestine Regions**

Region	Main Functions	Key Digestive Actions	Key Absorptive Actions
Duodenum	Chemical digestion	Receives bile & pancreatic enzymes; neutralises chyme	Starts absorption of sugars, amino acids, small fatty acids
Jejunum	Nutrient absorption	Completes digestion with brush-border enzymes	Absorbs carbohydrates, proteins, fats, vitamins, minerals
Ileum	Final absorption & immune defence	Minimal digestion; prepares residue for large intestine	Absorbs bile salts, vitamin B_{12}, remaining nutrients; immune surveillance

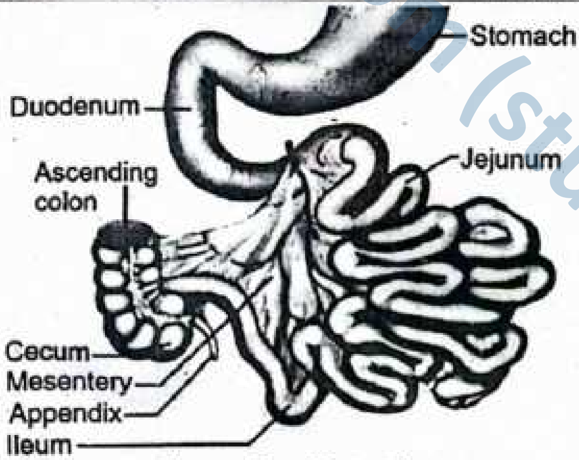

Figure: Small Intestine

SLOs Based (MCQs)

SMALL INTESTINE

1. **Which hormone stimulates the secretion of gastric juice by acting on parietal cells?**
   A. Secretin B. Gastrin
   C. Cholecystokinin D. Insulin

2. **Which enzyme activates trypsinogen into trypsin in the small intestine?**
   A. Pepsin B. Amylase
   C. Enterokinase D. Lipase

3. **What is the primary function of bile in digestion?**
A. Hydrolyse fats
B. Emulsify fats
C. Neutralize stomach acid
D. Activate trypsinogen

4. **Which part of the small intestine is primarily responsible for nutrient absorption?**
A. Duodenum B. Jejunum
C. Ileum D. Colon

5. **What structure increases the surface area for absorption in the small intestine?**
A. Gastric pits
B. Rugae
C. Villi and microvilli
D. Crypts of Lieberkühn

6. **Which enzyme in pancreatic juice breaks down starch?**
A. Trypsin B. Lipase
C. Pancreatic amylase D. Pepsin

7. **Chylomicrons are absorbed into:**
A. Blood capillaries B. Hepatic portal vein
C. Lymphatic lacteals D. Stomach wall

Answer Key

1. B	2. C	3. B	4. C	5. C	6. C	7. C

Short Questions

Q.1 Identify the major regions of the small intestine and explain their roles.

Ans. The small intestine is divided into three regions:

1. **Duodenum** – the first, shortest segment; receives **chyme** from the stomach along with bile from the liver/gallbladder and pancreatic juice from the pancreas. It initiates chemical digestion by mixing chyme with digestive secretions.
2. **Jejunum** – the middle section; has a thick wall and abundant villi for maximum nutrient absorption. Most enzymatic digestion and absorption of carbohydrates, proteins, and lipids occurs here.
3. **Ileum** – the final, longest section; absorbs vitamin B_{12}, bile salts, and remaining nutrients. It ends at the ileocecal valve, controlling entry into the large intestine.

Together, these regions ensure efficient digestion and absorption.

MARKING SCHEME (3 Marks)
- Duodenum and its role (1.0)
- Jejunum and its role (1.0)
- Ileum and its role (1.0)

Q.2 Describe the function of pancreatic juice and list its enzymes.

Ans. Pancreatic juice, secreted by the pancreas into the duodenum, is **alkaline** due to its bicarbonate content. This alkalinity **neutralises acidic chyme** from the stomach, creating an optimal pH for intestinal and pancreatic enzymes. Key enzymes include:

- **Pancreatic amylase** – breaks starch into maltose and smaller carbohydrates.
- **Pancreatic lipase** – digests triglycerides into fatty acids and glycerol.
- **Trypsinogen** and **chymotrypsinogen** – inactive protease precursors that, when activated, break proteins into peptides.
- By providing the correct pH and enzymes, pancreatic juice supports efficient digestion of all macronutrients.

MARKING SCHEME (3 Marks)
- Effect on pH (1.0)
- Names of enzymes (1.0)
- Functions of enzymes (1.0)

Q.3 Explain how trypsin and chymotrypsin are activated.

Ans. Trypsin is secreted by the pancreas as inactive **trypsinogen**. In the small intestine, **enterokinase** (enteropeptidase), secreted by intestinal glands, converts trypsinogen to active **trypsin**. Trypsin then acts autocatalytically to activate more trypsinogen and also converts **chymotrypsinogen** into active **chymotrypsin**. Both enzymes are proteases that break proteins into smaller polypeptides and peptides. This cascade ensures that these powerful enzymes are only activated in the intestinal lumen, preventing self-digestion of pancreatic tissue.

MARKING SCHEME (3 Marks)
- Trypsin activation by enterokinase (1.5)
- Chymotrypsin activation by trypsin (1.5)

Q.4 Describe how bile aids lipid digestion.

Ans. *Bile is* a yellow-green fluid produced by the **liver** and stored in the **gallbladder**. It contains bile salts that **emulsify fats**, breaking large fat globules into tiny droplets. This mechanical process increases the **surface area available for pancreatic lipase to act upon, greatly accelerating**

MARKING SCHEME (3 Marks)
- Definition of emulsification (1.0)
- Role in increasing surface area for lipase (2.0)

fat digestion. Emulsification is **not** enzymatic; it simply prepares lipids for efficient enzymatic breakdown into fatty acids and glycerol. Bile also helps with the absorption of fat-soluble vitamins (A, D, E, K).

Q.5 Explain how carbohydrates are absorbed in the small intestine.

Ans. Carbohydrate digestion produces **monosaccharides**, mainly glucose, along with fructose and galactose. Glucose and galactose are absorbed into intestinal epithelial cells via **secondary active transport** using sodium-glucose transport proteins (SGLT). Fructose is absorbed by facilitated diffusion. From epithelial cells, monosaccharides diffuse into the capillaries within each villus. They are transported via the **hepatic portal vein** to the liver, where glucose can be stored as glycogen or released into the bloodstream. This process ensures rapid delivery of carbohydrate energy to body tissues.

MARKING SCHEME (3 Marks)
- Name of smallest product (1.0)
- Transport mechanism (1.0)
- Pathway via hepatic portal vein (1.0)

Q.6 Explain how lipids are absorbed in the small intestine.

Ans. Lipids are digested into **fatty acids** and **monoglycerides**. These products enter intestinal epithelial cells by **simple diffusion** after being packaged into micelles with bile salts. Inside the cells, fatty acids and monoglycerides are reassembled into **triglycerides** and combined with proteins to form **chylomicrons**. These chylomicrons enter the **lacteals** (lymphatic vessels) within villi, travel through the lymphatic system, and eventually enter the bloodstream via the thoracic duct. This pathway bypasses the liver initially, delivering lipids directly into circulation.

MARKING SCHEME (3 Marks)
- Name of smallest product (1.0)
- Transport mechanism (1.0)
- Lymphatic pathway (1.0)

Q.7 Describe the structure and function of villi in the small intestine.

Ans. Villi are **finger-like projections** of the intestinal mucosa that greatly increase surface area for nutrient absorption. Each villus is covered by simple columnar epithelial cells with **microvilli** on their apical surfaces, forming a brush border. Internally, a villus contains a **capillary network** for absorption of carbohydrates and proteins, and a **lacteal** for lipid transport. The thin epithelial barrier allows rapid diffusion and transport of nutrients into blood and lymph. This structural specialisation maximises absorption efficiency, enabling the small intestine to process large volumes of digested material.

MARKING SCHEME (3 Marks)
- Structure, including microvilli and internal components (1.5)
- Function in nutrient absorption (1.5)

SLO [B-12-R-29]
Trace the absorption of digested products from the small intestine lumen to the blood capillaries and lacteals of the villi.

ABSORPTION PATHWAY OF DIGESTED PRODUCTS IN SMALL INTESTINE

After digestion in the small intestine, the resulting **monomers and small molecules** must be transferred from the **lumen** into the **circulatory systems**. The process involves crossing the intestinal wall via the specialised structures of the **villi**.

1. Structure of the Villi

Villi are **microscopic, finger-like projections** arising from the mucosal surface of the small intestine, dramatically increasing the absorptive surface area. Each villus is lined by a **single layer of columnar epithelial cells**, many of which bear **dense microvilli**, forming the *brush border*. This arrangement maximizes nutrient absorption by bringing digestive enzymes and transport proteins into close contact with chyme.

Within each villus lies a specialized internal structure:

- **Blood capillary network** – a rich mesh of capillaries drains into the *hepatic portal vein*, transporting water-soluble nutrients such as monosaccharides, amino acids, water-soluble vitamins, and minerals directly to the liver for processing.
- **Central lacteal** – a lymphatic vessel responsible for the transport of absorbed lipids and lipid-soluble vitamins (A, D, E, K) as chylomicrons via the lymphatic system into systemic circulation.
- **Goblet cells** – interspersed among absorptive cells, secreting mucus to lubricate the villus surface and facilitate smooth chyme passage.

- **Smooth muscle fibers** – present in the core of the villus, gently contract to aid in the mixing of intestinal contents and to move absorbed materials into the capillaries and lacteals.

This intricate structural organization enables villi to perform **efficient, selective, and continuous nutrient absorption** throughout digestion.

2. Pathway for Water-Soluble Nutrients (Carbohydrates, Proteins, Water-Soluble Vitamins)

The absorption and transport of water-soluble nutrients in the small intestine involve a series of specialized steps designed to ensure rapid and efficient delivery to the liver:

1. **Digested Form in the Intestinal Lumen**
    - **Carbohydrates** are enzymatically broken down into *monosaccharides* such as glucose, galactose, and fructose.
    - **Proteins** are hydrolyzed into their building blocks, *amino acids*.
    - **Water-soluble vitamins** (e.g., vitamin C and B-complex vitamins) remain in a dissolved state, ready for absorption.
2. **Absorption into Enterocytes**

    These nutrients cross the brush-border membrane of the columnar epithelial cells:
    - **Active transport** is used for glucose, galactose, and most amino acids, requiring carrier proteins and energy (ATP).
    - **Facilitated diffusion** allows fructose and certain vitamins to pass along their concentration gradients without energy expenditure.
3. **Movement into Blood Capillaries**

    Inside the villus, absorbed nutrients exit the enterocytes through the basolateral membrane and diffuse into the **dense capillary network**, aided by a short diffusion distance and rich blood supply.
4. **Transport to the Liver**

    The capillaries of the villi merge into progressively larger veins, forming the **hepatic portal vein**, which delivers nutrient-rich blood directly to the liver. Here, the liver regulates nutrient levels, stores excess glucose as glycogen, synthesizes plasma proteins, and releases nutrients into the systemic circulation as required.

This direct **lumen-to-liver route** ensures that water-soluble nutrients are processed before entering the general circulation, maintaining metabolic stability.

3. Pathway for Lipid-Soluble Nutrients (Fats, Fat-Soluble Vitamins A, D, E, K)

The absorption and transport of lipid-soluble nutrients involve a complex pathway distinct from that of water-soluble substances, reflecting their hydrophobic nature:

1. **Digested Form in the Intestinal Lumen**

    Dietary triglycerides are broken down by **pancreatic lipase** into their constituent **fatty acids** and **monoglycerides**. These smaller lipid molecules remain emulsified by bile salts, forming micelles that facilitate their movement toward the intestinal epithelial surface.
2. **Absorption into Enterocytes**

    Fatty acids and monoglycerides **diffuse passively** across the phospholipid bilayer of the enterocyte's brush border membrane due to their lipid solubility. Inside the epithelial cells, these components are reassembled into **triglycerides** within the smooth endoplasmic reticulum.
3. **Formation of Chylomicrons**

    The reformed triglycerides are packaged with **phospholipids, cholesterol, and specific proteins (apolipoproteins)** to form **chylomicrons**, which are large, lipoprotein particles optimized for lipid transport in an aqueous environment.
4. **Entry into Lacteals**

    Chylomicrons are too large to enter blood capillaries directly. Instead, they enter the **lacteals**, specialized lymphatic vessels within the core of each villus, initiating their journey through the lymphatic system.
5. **Transport to the Bloodstream**

    The lymphatic fluid containing chylomicrons travels through progressively larger lymph vessels and eventually drains into the **thoracic duct**, which empties into the **left subclavian vein** near the heart. This route bypasses the liver initially, allowing lipids direct access to systemic circulation for use or storage by peripheral tissues.

This pathway ensures efficient absorption and systemic distribution of lipids and fat-soluble vitamins, critical for energy metabolism and cellular functions.

4. Absorption of Water and Minerals

The small intestine is not only responsible for nutrient absorption but also plays a critical role in the uptake of water and essential minerals, maintaining the body's fluid and electrolyte balance.

- **Water Absorption:**

  Water is absorbed primarily through **osmosis**, a passive movement driven by the osmotic gradient created by nutrient absorption. As solutes like glucose, amino acids, and electrolytes enter the intestinal epithelial cells, water follows from the lumen into the cells and then into the bloodstream. This process occurs throughout the small intestine, ensuring efficient rehydration and maintaining blood volume and tissue hydration.

- **Mineral Absorption:**

  Minerals are absorbed via active and passive transport mechanisms depending on the mineral and region of the intestine:

  - **Calcium and Iron:**

    These are mainly absorbed in the **duodenum and jejunum**. Calcium absorption is tightly regulated by vitamin D, which promotes active transport via calcium-binding proteins. Iron is absorbed as ferrous ions through specialized transporters and stored or transported bound to *proteins* such as ferritin or transferrin.

  - **Sodium, Potassium, and Chloride:**

    These electrolytes are absorbed throughout the small intestine. Sodium absorption often occurs via active transport, frequently coupled with glucose or amino acid uptake, while potassium and chloride absorption involve both passive diffusion and active transport mechanisms to maintain ionic balance and proper cellular function.

Together, water and mineral absorption in the small intestine supports fluid homeostasis, nerve conduction, muscle contraction, and overall metabolic activity essential for health.

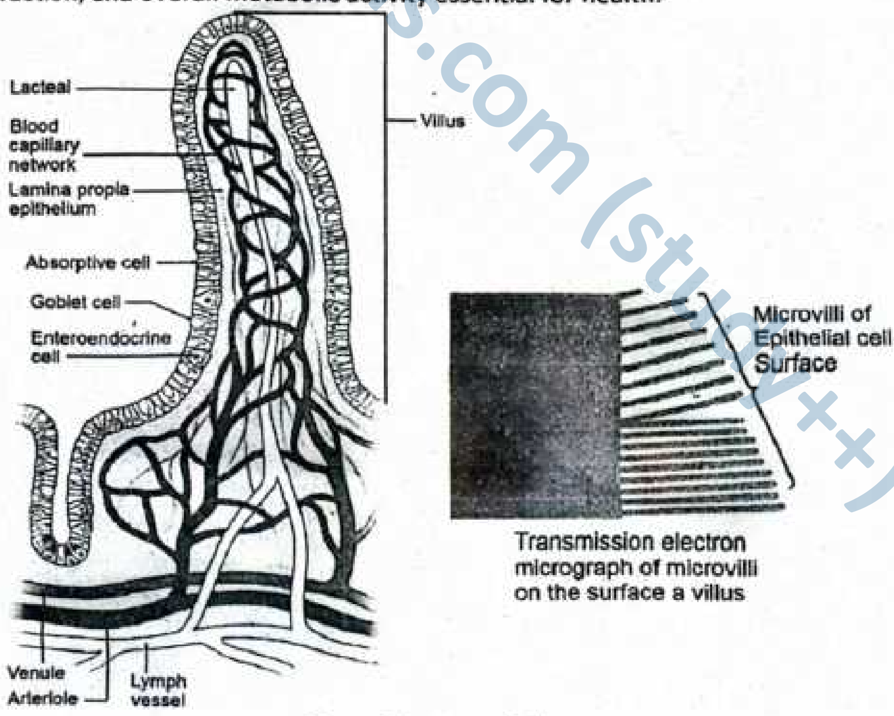

Figure: Structure of villus

- **Absorption Routes**

Nutrient Type	Entry into Epithelial Cell	Transport Route	First Destination
Monosaccharides	Active/facilitated transport	Blood capillaries	Liver (via portal vein)
Amino acids	Active transport	Blood capillaries	Liver (via portal vein)
Water-soluble vitamins	Active/facilitated transport	Blood capillaries	Liver
Lipids (fatty acids, monoglycerides)	Diffusion → chylomicrons	Lacteals (lymphatics)	Systemic circulation
Fat-soluble vitamins (A, D, E, K)	Diffusion → chylomicrons	Lacteals (lymphatics)	Systemic circulation
Minerals (Ca^{2+}, Fe^{2+})	Active transport	Blood capillaries	Liver

- **Comparison of Digestion in Stomach and Small Intestine**

Feature	Stomach	Small Intestine
Site	Located between oesophagus and duodenum	Extends from duodenum to ileum
Main Digestive Enzymes	Pepsin (secreted as pepsinogen)	Maltase, sucrase, lactase, peptidases, lipase, amylase (from pancreas)
Enzyme Activation	Pepsinogen activated by HCl to form pepsin	Enzymes are either active or activated by intestinal or pancreatic secretions
pH Conditions	Acidic (pH ~1.5–2.0) due to hydrochloric acid	Alkaline (pH ~7.5–8.5) due to bicarbonate from pancreas and bile
Substrates Digested	Proteins (partial digestion)	Carbohydrates, proteins (further), lipids
Mechanical Activity	Churning by muscular walls	Segmentation and peristalsis
Digestive Secretions	Gastric juice (HCl, pepsinogen, mucus)	Intestinal juice, pancreatic juice, and bile
Hormonal Regulation	Gastrin stimulates gastric juice secretion	Secretin and CCK stimulate pancreatic juice and bile release
Absorption	Limited (e.g., alcohol, some drugs)	Extensive absorption of digested nutrients
Digestive Product	**Chyme** (partially digested, acidic mixture of food and gastric secretions)	**Chyle** (milky fluid rich in emulsified fats) and fully digested nutrients
Role in Digestion	Initiates protein digestion; mixes food into chyme	Completes digestion and facilitates nutrient absorption

SLO [B-12-R-30]
Describe the component parts of the large intestine with their respective roles.

COMPONENT PARTS OF THE LARGE INTESTINE AND THEIR ROLES

The **large intestine** is the final section of the digestive tract, approximately 1.5 meters long, responsible primarily for water absorption, formation of feces, and elimination of waste. It is divided into several distinct parts, each with specialized functions:

Cecum

- **Location and Structure**

The **cecum** is a ***pouch-like, blind***-ended sac that forms the **beginning of the large intestine.** It is situated in the ***lower right quadrant*** **of the abdomen**, where the small intestine joins the large intestine. The cecum is relatively wide compared to other parts of the intestine and serves as a key junction point in digestion.

- **Connection with the Ileum and Ileocecal Valve**

  The terminal portion of the small intestine, called the **ileum**, empties its contents into the cecum through the **ileocecal valve**. This valve is a one-way muscular sphincter that **regulates the passage** of chyme from the ileum into the large intestine, preventing the backflow of fecal material into the small intestine. This mechanism helps maintain a proper balance of microbial populations and protects the small intestine from potential contamination.

- **Functions of the Cecum**

  Although the cecum has a **minor role in nutrient absorption**, it absorbs some residual **water and electrolytes** such as sodium and chloride. Its major function is to act as a **fermentation chamber** where beneficial bacteria break down any remaining undigested carbohydrates and fibers. This bacterial fermentation produces gases like carbon dioxide and methane, as well as **short-chain fatty acids** which can be absorbed and provide additional energy to the body.

- **Immune Role**

  The cecum also contains **lymphoid tissue** that contributes to immune defense by monitoring intestinal bacteria and helping protect the body from harmful pathogens.

Colon

- **Structure**

  The **colon** is the longest section of the large intestine, measuring about 1.5 meters in length. It follows the cecum and extends to the rectum. The colon is divided into four distinct regions, each with specific anatomical positions and roles:
  - **Ascending colon** (travels upward on the right side of the abdomen)
  - **Transverse colon** (crosses the abdomen horizontally from right to left)
  - **Descending colon** (runs downward on the left side)
  - **Sigmoid colon** (S-shaped section leading into the rectum)

- **Absorption of Water and Electrolytes**

  The primary function of the colon is to **absorb water and electrolytes** such as sodium and chloride from the remaining indigestible food residue, known as chyme. As water is absorbed, the chyme gradually transforms from a watery liquid into a more solid mass called **feces**. This absorption process is vital to maintain the body's fluid and electrolyte balance and prevent dehydration.

- **Role of Gut Microbiota**

  The colon is also home to a vast community of **beneficial bacteria** that play an essential role in digestion. These bacteria **ferment undigested dietary fibers**, producing gases like carbon dioxide, methane, and hydrogen. Additionally, this fermentation process synthesizes important vitamins such as **vitamin K** and some **B-complex vitamins**, which are absorbed by the colon and contribute to overall nutrition.

- **Other Functions**

  Besides absorption and fermentation, the colon compacts fecal matter and stores it until it is ready to be eliminated. The colon's muscular walls also perform slow, rhythmic contractions called **haustral churning** and **mass movements**, which aid in mixing the contents and propelling feces toward the rectum.

Rectum

- **Location and Structure**

  The **rectum** is the final segment of the large intestine, measuring about **12 to 15 centimeters in length**. It lies just before the anal canal and serves as the **terminal storage site** for feces within the digestive system. The rectum follows the sigmoid colon and is positioned in the pelvis, conforming to the curvature of the sacrum.

- **Function as a Storage Site**

  The primary role of the rectum is to **temporarily store fecal matter** until it is ready to be expelled. The walls of the rectum are highly stretchable and contain numerous stretch receptors that detect when the rectum becomes distended with feces. This distension sends signals to the nervous system, initiating the **urge to defecate.**

- **Sensory and Motor Coordination**

  Once the rectum is full, these stretch receptors activate a coordinated response involving voluntary and involuntary muscle actions. The internal anal sphincter (involuntary muscle) relaxes reflexively, while the external anal sphincter (under voluntary control) allows conscious control over the timing of defecation. This system provides a balance between the automatic elimination of waste and conscious regulation.

Anal Canal and Anus

Structure of the Anal Canal

The **anal canal** is the final segment of the digestive tract, approximately 3 to 4 centimeters long, connecting the rectum to the external opening of the body, the **anus**. It is lined with specialized mucosa that transitions from the columnar epithelium of the rectum to the stratified squamous epithelium near the anus, which is better suited to withstand abrasion during defecation.

Sphincters Controlling Defecation

The anal canal contains two critical muscular rings called **sphincters** that regulate the passage of feces:

- **Internal Anal Sphincter:** This is an involuntary smooth muscle ring that remains tonically contracted to maintain continence. It relaxes reflexively when the rectum is full, allowing feces to move toward the anus.
- **External Anal Sphincter:** This is a voluntary skeletal muscle ring surrounding the internal sphincter. It provides conscious control over defecation, allowing a person to delay stool passage until an appropriate time and place.

Function in Waste Elimination

Together, these sphincters maintain fecal continence and coordinate the controlled release of feces during defecation. Sensory nerves in the anal canal detect stool consistency and pressure, helping distinguish between gas, liquid, and solid waste to prevent accidental release.

Together, these parts of the large intestine coordinate to complete the digestive process by reclaiming water, storing and compacting waste, and facilitating its controlled elimination.

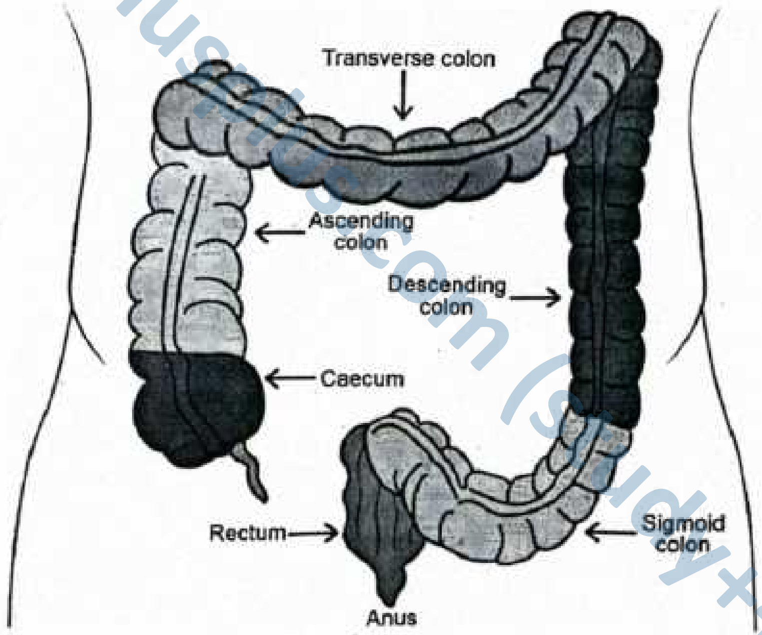

Figure: Large Intestine

For Your Information!

The appendix contains a small amount of mucus associated lymphoid tissue which gives the appendix an undetermined role in immunity. However, the appendix is known to be *important in foetal or fetal* life as it contains endocrine cells that release biogenic amines and peptide hormones important for homeostasis for during early growth and development. Appendicitis is an inflammation of the vermiform appendix and usually occurs because of obstruction of the appendix. The removal of appendix is called appendectomy.

Functions of the Large Intestine

The large intestine performs several essential functions, primarily related to the absorption and processing of waste:

1. **Water and Electrolyte Absorption**

   Although no further enzymatic digestion occurs, the proximal region of the large intestine reabsorbs water and electrolytes. This compacts the waste material into **faeces**, which are stored temporarily in the rectum.

2. **Vitamin Absorption**

   Symbiotic bacteria residing in the colon synthesize vitamins, notably **Vitamin K** and **Biotin (Vitamin B7)**. These are absorbed through the intestinal wall and contribute to overall nutritional balance.

3. **Reduction of Acidity and Protection Against Infection**

   The mucosa of the large intestine secretes **bicarbonate ions** to neutralize acidic waste products and functions as a **mucosal barrier** to prevent microbial invasion.

SLOs Based (MCQs)

LARGE INTESTINE AND LIVER

1. **Which structure marks the beginning of the large intestine?**
   A. Rectum B. Duodenum
   C. Caecum D. Jejunum

2. **Which vitamin is primarily absorbed in the large intestine due to bacterial synthesis?**
   A. Vitamin D B. Vitamin A
   C. Vitamin C D. Vitamin K

3. **What is the main function of bile salts?**
   A. Protein digestion
   B. Carbohydrate emulsification
   C. Fat emulsification D. Vitamin storage

4. **What causes the brown colour of faeces?**
   A. Urea B. Bilirubin metabolites
   C. Trypsin D. Glycogen

5. **Which enzyme activates trypsinogen in the small intestine?**
   A. Trypsin B. Enterokinase
   C. Pepsin D. Amylase

6. **Which pancreatic enzyme digests carbohydrates?**
   A. Lipase B. Trypsin
   C. Amylase D. Nuclease

7. **What is the role of the liver in detoxification?**
   A. It stores toxins
   B. It metabolizes toxins into non-toxic forms
   C. It prevents toxin entry into the bloodstream
   D. It converts toxins into glucose

8. **What is the function of bicarbonates in pancreatic juice?**
   A. To digest proteins B. To activate enzymes
   C. To neutralize stomach acid
   D. To emulsify fats

9. **Which of the following is NOT a function of the liver?**
   A. Bile production B. Vitamin C synthesis
   C. Detoxification
   D. Regulation of blood glucose

10. **What is the function of the external anal sphincter?**
    A. Automatic expulsion of faeces
    B. Enzymatic secretion
    C. Voluntary control of defaecation
    D. Absorption of water

Answer Key

1. C	2. D	3. C	4. B	5. B	6. C	7. B	8. C	9. B	10. C

Short Questions

Q.1 Locate and explain the role of the ileocecal sphincter in the digestive system.

MARKING SCHEME (3 Marks)
• Location at ileum–caecum junction (1.0)
• Regulation of chyme flow (1.0)
• Prevention of backflow (1.0)

Ans. The **ileocecal sphincter** is located at the junction between the **ileum** (last segment of the small intestine) and the **caecum** (first part of the large intestine). *Structurally, it is a circular band of smooth muscle that regulates the passage of digested material.* Its main function is to **control the flow of chyme** from the small intestine into the large intestine, preventing premature entry that could disrupt absorption. Additionally, it **prevents backflow** of colonic contents, protecting the small intestine from bacterial contamination.

The opening and closing of this sphincter are controlled by reflexes involving the enteric nervous system, responding to pressure and chemical composition of intestinal contents. By coordinating transit, it ensures proper absorption in the ileum before waste passes into the large intestine.

Q.2 Describe the structure and function of the vermiform appendix.

3 **Ans.** The **vermiform appendix** is a narrow, blind-ended tube measuring approximately **9 cm** in length, attached to the posteromedial wall of the caecum just below the ileocecal junction. It has a lumen that opens into the caecum and a wall rich in **lymphoid tissue**, which contributes to immune function. Structurally, it shares the same layers as the intestinal wall: mucosa, submucosa, muscularis, and serosa. Functionally, the appendix is thought to act as a **reservoir for beneficial gut bacteria** and participates in immune surveillance, particularly in early life, by producing lymphocytes. While its digestive role is minimal, it plays a supportive part in maintaining intestinal microbial balance and mucosal immunity.

> **MARKING SCHEME (3 Marks)**
> - Structure and dimensions (0.75)
> - Lymphoid tissue presence (0.75)
> - Function in immunity / microbial balance (1.5)

Q.3 List the main functions of the large intestine.

Ans. The large intestine performs several essential roles:

1. **Absorption of water and electrolytes**, converting liquid chyme into semi-solid faeces.
2. **Absorption of vitamins** produced by gut microbiota, particularly vitamin K and biotin.
3. **Secretion of mucus and bicarbonate**, which protect the mucosa from mechanical damage and acidic fermentation products.
4. **Formation and storage of faeces** before elimination.
5. **Housing beneficial bacteria** that assist in fermentation of undigested carbohydrates.

These functions maintain fluid balance, support nutrition through vitamin absorption, and safeguard intestinal health.

> **MARKING SCHEME (3 Marks)**
> - Any three distinct functions (1.0 each)

Q.4 Explain how the large intestine assists in vitamin absorption.

Ans. Vitamin absorption in the large intestine is largely dependent on the activity of **resident bacteria**. These beneficial microbes synthesise vitamins, most notably **vitamin K** and **biotin**, during the fermentation of undigested carbohydrates. Once produced, these vitamins are absorbed through the mucosal lining of the large intestine into the bloodstream. This process helps supplement dietary intake, ensuring adequate supply for functions such as blood clotting (vitamin K) and metabolic enzyme activity (biotin). The symbiotic relationship between gut flora and the host highlights the importance of a balanced microbiome for optimal nutrient absorption.

> **MARKING SCHEME (3 Marks)**
> - Role of bacteria in synthesis (2.0)
> - Naming the vitamin(s) (1.0)

Q.5 Describe the mechanism that triggers the defaecation reflex in adults.

Ans. The defaecation reflex begins when the **rectal wall** is distended by accumulating faeces, activating **stretch receptors**. These receptors send signals to the spinal cord, initiating reflexive **peristaltic contractions** of the rectal muscles. The **internal anal sphincter** (involuntary smooth muscle) relaxes, allowing faeces to move toward the anal canal. The **external anal sphincter** (voluntary skeletal muscle) remains contracted until conscious relaxation permits elimination. This coordinated action between reflexive and voluntary control ensures controlled bowel movements. *Inhibition* of the reflex allows temporary faecal storage until socially appropriate conditions for defaecation are met.

> **MARKING SCHEME (3 Marks)**
> - Role of rectal distension/stretch receptors (2.0)
> - Role of sphincters (1.0)

Q.6 Explain how bile is involved in digestion.

Ans. Bile is a digestive fluid produced by the liver and stored in the **gallbladder**, containing **bile salts, bile pigments (bilirubin), cholesterol, and water**. Its primary digestive function is to **emulsify fats**, breaking large lipid droplets into smaller ones. This increases surface area for pancreatic lipase action, enhancing fat digestion. Bile also aids in the **absorption of fatty acids, monoglycerides, and fat-soluble vitamins (A, D, E, K)** in the small intestine.

> **MARKING SCHEME (3 Marks)**
> - Bile composition (1.0)
> - Fat emulsification (1.0)
> - Absorption facilitation (1.0)

Additionally, bile pigments, especially bilirubin, contribute to the brown colour of faeces after bacterial breakdown. Although bile contains no enzymes, its role is essential for efficient lipid processing.

Q.7 State the major functions of the liver.

Ans. The liver is a multifunctional organ performing over **500 physiological tasks**, including: 3/3

1. **Bile production** for fat digestion.
2. **Detoxification** of harmful substances such as drugs and alcohol.
3. **Glucose metabolism** via glycogenesis, glycogenolysis, and gluconeogenesis.
4. **Protein synthesis**, including plasma proteins like albumin and clotting factors.
5. **Urea formation** from ammonia detoxification.
6. **Immune function** through Kupffer cells that filter pathogens.

These functions are vital for digestion, metabolism, detoxification, and immunity.

> **MARKING SCHEME (3 Marks)**
> - Any three functions (1.0 each)

Q.8 Identify and describe the role of pancreatic enzymes in digestion.

Ans. The pancreas secretes a range of digestive enzymes into the small intestine:

- **Amylase** – breaks down starch into maltose and other small carbohydrates.
- **Lipase** – digests triglycerides into fatty acids and monoglycerides.
- **Proteases** (trypsin, chymotrypsin, carboxypeptidase) – hydrolyse proteins into peptides and amino acids.
- **Nucleases** – degrade DNA and RNA into nucleotides.

These enzymes work synergistically to break down macronutrients into absorbable units, enabling efficient nutrient uptake in the small intestine.

> **MARKING SCHEME (3 Marks)**
> - Names of enzymes (1.0)
> - Functions in macronutrient digestion (2.0)

Q.9 Explain how trypsinogen is activated in the small intestine.

Ans. Trypsinogen, an inactive protease precursor secreted by pancreatic acinar cells, is converted to **trypsin** by the enzyme **enterokinase** (enteropeptidase) found in the brush border of the duodenum. Active trypsin then activates additional trypsinogen molecules and other zymogens such as **chymotrypsinogen** and **procarboxypeptidase**. This cascade ensures that proteolytic enzymes are only active in the intestinal lumen, preventing autodigestion of the pancreas. The activation of trypsinogen is thus critical for initiating protein digestion in the small intestine.

> **MARKING SCHEME (3 Marks)**
> - Enterokinase activation (1.0)
> - Trypsin autocatalysis (1.0)
> - Activation of other enzymes (1.0)

Q.10 Explain how bicarbonate from pancreatic juice aids digestion.

Ans. Bicarbonate ions in pancreatic juice, secreted by ductal cells, neutralise the **acidic chyme** entering the duodenum from the stomach, raising the pH to around **7–8**. This alkaline environment:

1. Protects the intestinal mucosa from acid damage.
2. Provides optimal conditions for pancreatic enzymes, such as amylase, lipase, and proteases, which function best in neutral to slightly alkaline pH.
3. By maintaining the correct pH balance, bicarbonate ensures efficient digestion and absorption of nutrients in the small intestine.

> **MARKING SCHEME (3 Marks)**
> - Neutralisation and pH increase (1.0)
> - Protection of mucosa (1.0)
> - Optimal enzyme function (1.0)

SLO [B-12-R-31]
Correlate the involuntary reflex for egestion in infants and the voluntary control in adults.

INVOLUNTARY REFLEX FOR EGESTION IN INFANTS AND VOLUNTARY CONTROL IN ADULTS

▶ Egestion and its Importance

Egestion is the process of eliminating undigested waste materials and feces from the body through the anus. This is the final step of digestion and is crucial for maintaining health by removing materials the body cannot use.

- **Involuntary Egestion Reflex in Infants**

  In **newborn infants,** the control of egestion is primarily **involuntary**. This means that the muscles responsible for expelling feces contract automatically without conscious effort. The reflex is triggered when the rectum becomes filled with feces, stimulating **stretch receptors** in the rectal walls. These receptors send signals to the spinal cord and brainstem, which then coordinate the relaxation of the internal anal sphincter and contraction of the rectal muscles to push feces out.

  Because the **external anal sphincter** in infants is **not yet fully developed** or under voluntary control, infants cannot consciously delay or control defecation. This involuntary reflex ensures that waste is expelled promptly, which is why infants do not have toilet control and pass stools frequently.

- **Development of Voluntary Control in Adults**

  As a child grows, the nervous system matures, and the **external anal sphincter**, composed of skeletal muscle, gains **voluntary control**. This development typically occurs between the ages of 18 months and 3 years and is essential for **toilet training**.

  In adults, egestion becomes a **voluntary action** involving both the involuntary reflex and conscious control mechanisms:
  - When the rectum fills, stretch receptors still initiate the reflex urge to defecate by relaxing the internal anal sphincter.
  - However, adults can consciously contract the external anal sphincter to **delay defecation** until an appropriate time and place.
  - When ready, relaxation of the external sphincter along with abdominal muscle contraction and increased intra-abdominal pressure facilitates feces expulsion.

- **Neural Coordination and Control**

  The shift from involuntary to voluntary control involves complex **neural pathways:**
  - The **enteric nervous system** manages local reflexes in the gut.
  - The **central nervous system** (brain and spinal cord) develops the ability to override reflexes, enabling conscious control over defecation.

  This balance between reflex action and voluntary control allows adults to regulate bowel movements, *contributing* to social appropriateness and personal hygiene.

  In summary, infants rely solely on an **involuntary reflex** for egestion due to undeveloped voluntary muscles and nervous control. As development progresses, the **external anal sphincter gains voluntary control,** allowing adults to consciously regulate defecation. This transition reflects the maturation of both muscular and neural systems essential for independent living.

SLO [B-12-R-32]
Explain the storage and metabolic role of liver.

STORAGE AND METABOLIC ROLE OF THE LIVER

The **liver** is the largest internal organ in the human body and plays a vital role in maintaining overall metabolism and homeostasis. Located in the upper right portion of the abdomen, it performs numerous functions essential for digestion, detoxification, and nutrient management.

Storage Functions of the Liver

The liver serves as the body's **central storage depot** for key nutrients, ensuring that cells have a steady supply of energy and essential substances, even between meals. This storage function allows the body to maintain **homeostasis** despite variations in dietary intake.

1. **Glycogen Storage and Glucose Regulation**
   - **Formation of Glycogen (Glycogenesis):**
     After a carbohydrate-rich meal, excess glucose in the blood is transported to the liver. Hepatocytes (liver cells) convert this surplus glucose into glycogen, a branched polysaccharide that is easily stored.
   - **Breakdown of Glycogen (Glycogenolysis):**
     When blood glucose levels drop, such as between meals or during exercise, the liver rapidly breaks down glycogen into glucose and releases it into the bloodstream.

- **Importance:**
  This mechanism ensures a constant glucose supply to vital organs, particularly the brain, which relies almost entirely on glucose for energy.

2. **Vitamin Storage**
   - **Fat-Soluble Vitamins (A, D, E, K):**
     These vitamins are absorbed from the digestive tract along with dietary fats and stored in the liver for long-term use.
     - **Vitamin A:** Required for vision, epithelial tissue maintenance, and growth.
     - **Vitamin D:** Essential for calcium absorption and bone health.
     - **Vitamin E:** Acts as an antioxidant, protecting cell membranes.
     - **Vitamin K:** Necessary for blood clotting factor synthesis.
   - **Vitamin B12 (Cobalamin):**
     Although water-soluble, B12 can be stored in the liver for several years, making it an exception among water-soluble vitamins. It is vital for red blood cell formation and nervous system function.

3. **Mineral Storage**
   - **Iron:** Stored mainly in the form of **ferritin,** a protein-iron complex. This stored iron is released into the blood when needed for **haemoglobin synthesis** during red blood cell production in the bone marrow.
   - **Copper:** Stored in trace amounts, copper is a cofactor in many enzymes involved in energy production, iron metabolism, and antioxidant defence.

4. **Physiological Importance of Liver Storage**
   The liver's storage function acts as a **nutrient bank,** protecting the body from fluctuations in dietary intake and preventing deficiencies. This is particularly critical:
   - During fasting or starvation.
   - In periods of rapid growth, pregnancy, or illness.
   - For tissues with high and constant energy demands, such as the brain and muscles.

Metabolic Functions of the Liver

The liver serves as the **primary metabolic control centre** of the body, regulating the chemical composition of the blood and coordinating the use, storage, and transformation of nutrients. Through its complex network of hepatocytes and enzymes, it ensures that carbohydrates, proteins, and fats are processed efficiently to meet the body's energy and structural needs.

1. **Carbohydrate Metabolism**
   - **Blood Glucose Regulation:**
     In addition to storing glycogen, the liver maintains stable blood glucose levels through two key processes:
     - **Glycogenolysis** – breakdown of stored glycogen into glucose during fasting or exercise.
     - **Gluconeogenesis** – synthesis of glucose from non-carbohydrate sources, such as amino acids, glycerol, and lactate, especially during prolonged fasting or intense activity.
   - **Physiological Role:**
     This constant regulation prevents both **hypoglycaemia** (low blood sugar) and **hyperglycaemia** (high blood sugar), ensuring a continuous energy supply for the brain and other vital organs.

2. **Protein Metabolism**
   - **Plasma Protein Synthesis:**
     The liver produces essential proteins, including:
     - **Albumin** – maintains osmotic pressure and transports hormones, fatty acids, and drugs.
     - **Clotting Factors** (e.g., fibrinogen, prothrombin) – required for normal blood coagulation.
   - **Amino Acid Processing:**
     The liver **deaminates** excess amino acids, removing their amino group and producing ammonia. Since ammonia *is toxic, the liver converts it into* **urea,** *which is then excreted by the kidneys.*
   - **Importance:**
     These processes maintain nitrogen balance and ensure that excess protein breakdown products do not accumulate in toxic levels.

3. **Lipid Metabolism**

- **Cholesterol and Lipoprotein Production:**
  The liver synthesises cholesterol and packages lipids into **lipoproteins** (VLDL, LDL, HDL) for transport to and from tissues.
- **Fatty Acid Breakdown:**
  Through **beta-oxidation**, fatty acids are broken down to produce ATP, supplying energy during fasting.
- **Triglyceride Regulation:**
  The liver also assembles and disassembles triglycerides, balancing fat storage and release.

4. **Detoxification and Biotransformation**

- **Toxin Removal:**
  The liver neutralises harmful substances such as drugs, alcohol, and environmental chemicals by converting them into water-soluble forms for excretion in urine or bile.
- **Endogenous Waste Processing:**
  It removes **bilirubin**, a breakdown product of haemoglobin, from the blood and excretes it into bile.
- **Metabolic Waste Handling:**
  Hormones and metabolic by-products are inactivated or broken down to prevent harmful accumulation.

5. **Central Role in Homeostasis**

By integrating carbohydrate, protein, and lipid metabolism while removing harmful substances, the liver maintains the **internal chemical balance** required for normal physiological function. This makes it a **vital organ for both nutrient processing and detoxification.**

In summary, the liver's **storage roles** ensure the body has reserves of glucose, vitamins, and minerals, while its **metabolic functions** manage vital biochemical pathways that sustain energy supply, protein synthesis, and detoxification. This makes the liver essential for maintaining the body's internal balance and overall health.

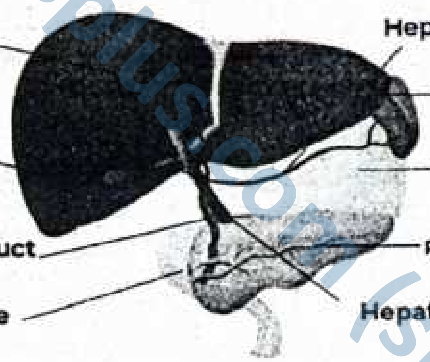

Figure: Liver

SLO [B-12-R-33]
Describe the composition of bile and relate the constituents with respective roles.

COMPOSITION OF BILE AND ROLES OF ITS CONSTITUENTS

Bile is a yellowish-green alkaline fluid produced continuously by the **liver** and stored in the **gallbladder** until needed. It is not an enzyme but a digestive secretion that plays a crucial role in **fat digestion** and the **elimination of waste products** from the body.

Composition of Bile and Their Roles

Constituent	Description	Role in the Body
Bile Salts (e.g., sodium glycocholate, sodium taurocholate)	Derived from bile acidn s, which are made from cholesterol.	*Emulsify fats* by breaking large fat globules into smaller droplets, increasing the surface area for lipase action. This enhances fat digestion and absorption of fat-soluble vitamins (A, D, E, K).

Bile Pigments (mainly **bilirubin and biliverdin)**	Waste products from the breakdown of hemoglobin in red blood cells.	Responsible for the yellow-green color of bile; excreted in feces, giving stool its characteristic brown color after conversion to stercobilin.
Cholesterol	A lipid molecule present in small amounts in bile.	Serves as a route for cholesterol excretion from the body.
Phospholipids (e.g., lecithin)	Fat-like molecules present alongside bile salts.	Help stabilize fat emulsions and prevent re-aggregation of fat droplets.
Electrolytes (sodium, potassium, calcium, chloride, bicarbonate ions)	Inorganic salts dissolved in bile.	Maintain bile's alkalinity (pH 7.6–8.6), which helps neutralize acidic chyme from the stomach.
Water	Makes up about 97% of bile.	Acts as the solvent for other constituents, aiding their transport and function.

Functional Significance

Bile is a complex digestive fluid produced by the liver and stored in the gallbladder. It plays a **dual role** in the digestive system—facilitating the breakdown and absorption of fats while also serving as a pathway for excreting certain waste products. Its unique chemical composition ensures that these functions are carried out effectively.

1. **Digestion of Lipids**
   - ***Emulsification:***
     Bile contains bile salts (derived from cholesterol) and phospholipids (mainly lecithin), which work together to break large fat globules into microscopic droplets. This process, called emulsification, increases the surface area of fats, making them more accessible to lipase enzymes secreted by the pancreas.
   - **Micelle Formation:**
     After lipase action, bile salts form small transport structures known as micelles, which carry the products of fat digestion—fatty acids, monoglycerides, cholesterol, and fat-soluble vitamins (A, D, E, K)—to the intestinal lining for absorption.
   - **Physiological Advantage:**
     Without bile salts, fat digestion would be inefficient, and most dietary lipids would pass unabsorbed into faeces.
2. **Waste Excretion**
   - **Bile Pigments:**
     The liver removes bilirubin, a yellow pigment produced from the breakdown of haemoglobin in red blood cells. This pigment is secreted into bile and eventually eliminated from the body in feces, giving them their characteristic brown colour.
   - **Cholesterol Removal:**
     Excess cholesterol, which cannot be broken down by the body, is excreted via bile. *Some* of this cholesterol is lost in feces, while the rest is reabsorbed and recycled.
   - **Detoxification Pathway:**
     Bile also acts as a route for eliminating lipid-soluble toxins and waste products, ensuring they do not accumulate in the body.
3. **pH Balance**
   - **Neutralisation of Gastric Acid:**
     The chyme entering the duodenum from the stomach is highly acidic due to gastric hydrochloric acid. Bile, being alkaline (pH ~7.5–8.5), helps neutralise this acidity.
   - **Protection of the Intestinal Mucosa:**
     This neutralisation prevents damage to the delicate epithelial lining of the small intestine.
   - **Optimisation of Enzyme Activity:**
     Pancreatic enzymes, such as lipase, amylase, and proteases, function best in a slightly alkaline environment. Bile thus creates ideal conditions for efficient digestion and nutrient absorption.

Through emulsification, micelle formation, waste excretion, and acid neutralisation, bile plays a **vital role** in digestion and homeostasis. It ensures that fats are digested efficiently, waste products are safely removed, and the intestinal environment remains suitable for enzymatic activity.

SLO [B-12-R-34]
Outline the structure of pancreas and explain its function as an exocrine gland.

STRUCTURE AND EXOCRINE FUNCTION OF THE PANCREAS

The pancreas is a soft, elongated gland situated in the upper abdomen, nestled between the stomach and the loop of the duodenum. It is both an **exocrine** and an **endocrine** gland, meaning it secretes substances through ducts as well as directly into the bloodstream. Here, we focus on its role as an **exocrine gland** in the digestive system.

1. Structure of the Pancreas

▶ Shape and Location

- The pancreas is about **15–20 cm long**, pale yellowish in colour, and has a lobulated appearance.
- It lies **horizontally across the posterior abdominal wall**, with its head nestled in the curve of the duodenum, its body extending across the midline, and its narrow tail reaching toward the spleen.

▶ External Features

- **Head:** The broadest part, positioned in the C-shaped curve of the duodenum.
- **Body:** Central elongated region lying behind the stomach.
- **Tail:** Narrow end close to the spleen.

▶ Internal Structure

- The pancreas is composed of two main tissue types:
  1. **Exocrine Tissue (Major Portion):** Made up of numerous secretory units called **acini**, which produce digestive enzymes.
  2. **Endocrine Tissue (Minor Portion):** Clusters of cells called **Islets of Langerhans**, responsible for hormone secretion (not the focus here).

▶ Duct System

- The small ducts from individual acini join to form larger ducts, eventually merging into the **main pancreatic duct**.
- This duct runs the length of the gland and joins with the **common bile duct**, opening into the duodenum at the **hepatopancreatic ampulla (ampulla of Vater)**.

2. Exocrine Function of the Pancreas

The exocrine pancreas is the major digestive gland of the small intestine. It produces **pancreatic juice**, a clear, alkaline secretion that is delivered through the pancreatic duct into the duodenum. This fluid plays a crucial role in the **chemical breakdown of nutrients** and in maintaining the **pH balance** required for enzymatic activity.

a) Composition of Pancreatic Juice

Pancreatic juice contains two main components — **digestive enzymes and bicarbonate ions** — each essential for specific digestive tasks.

1. **Digestive Enzymes**
   - **Amylase**
     - **Function:** Breaks down complex carbohydrates (starch) into maltose, a disaccharide.
     - **Importance:** Begins carbohydrate digestion in the small intestine after salivary amylase is inactivated by stomach acid.
   - **Lipase**
     - **Function:** Acts on triglycerides, splitting them into fatty acids and monoglycerides.
     - **Importance:** Works most efficiently after bile salts emulsify fats into small droplets.
   - **Proteolytic Enzymes** (secreted as inactive **zymogens** to prevent self-digestion of the pancreas):
     - **Trypsinogen** → Converted to trypsin by **enteropeptidase** in the duodenal lining; trypsin then activates other proteases.
     - **Chymotrypsinogen** → Activated by trypsin to chymotrypsin, which digests proteins into short peptides.

- Procarboxypeptidase → Activated to carboxypeptidase, which removes single amino acids from the ends of peptides.

2. **Bicarbonate Ions (HCO_3^-)**
   - **Function:** Neutralise the highly acidic chyme arriving from the stomach.
   - **Importance:** Raises the pH to about 7.8–8.4, which is the optimal range for pancreatic enzymes and prevents damage to the duodenal mucosa.

b) Regulation of Pancreatic Secretion

Pancreatic juice secretion is **precisely regulated** to match the type and quantity of food entering the small intestine. Two main control mechanisms operate:

1. **Nervous Control**
   - The **vagus nerve** stimulates the pancreas during the **cephalic phase** (sight, smell, or thought of food) and the **gastric phase** (when food is in the stomach), increasing both enzyme and bicarbonate secretion.
2. **Hormonal Control**
   - **Secretin**
     - Released by the duodenum in response to acidic chyme.
     - Stimulates the pancreas to release a bicarbonate-rich watery secretion to neutralise acidity.
   - **Cholecystokinin (CCK)**
     - Released by the duodenum in response to fats and proteins.
     - Stimulates secretion of enzyme-rich pancreatic juice for digestion.

c) Functional Importance

The exocrine pancreas ensures that the chemical breakdown of food is complete before absorption in the small intestine:

- **Complete Nutrient Digestion:**
  - Carbohydrates → Disaccharides (then broken down further by intestinal enzymes).
  - Proteins → Peptides and amino acids.
  - Fats → Fatty acids and monoglycerides.
- **Alkaline Environment:** Maintains optimal pH for enzyme action and protects the intestinal wall from gastric acid damage.
- **Coordination with Other Organs:** Works alongside bile from the liver and enzymes from the intestinal lining to make digestion efficient.

▶ **Exocrine Pancreas**

Feature	Description
Structural Unit	Acinus (cluster of secretory cells)
Duct	Main pancreatic duct joins bile duct
Secretion	Pancreatic juice (enzymes + bicarbonate)
Enzymes	Amylase, lipase, proteases
pH	Slightly alkaline (7.8–8.4)
Regulation	Vagus nerve, Secretin, CCK

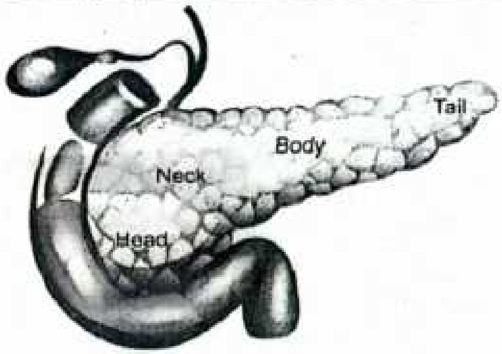

Figure: Structure of Pancreas

SLO [B-12-R-35]
Relate secretion of bile and pancreatic juice with the secretin of hormone.

SECRETION OF BILE AND PANCREATIC JUICE IN RELATION TO SECRETIN

Secretin is a peptide hormone produced by the **S cells** of the duodenal mucosa. Its primary role is to regulate the **alkaline environment** of the small intestine by stimulating the release of bicarbonate-rich fluids from accessory digestive glands.

1. Stimulus for Secretin Release

When a bolus of food leaves the stomach, it is mixed with gastric juice, forming **acidic chyme**. This chyme has a **low pH**, often between 2.0 and 4.0, due to the presence of hydrochloric acid. As the chyme passes through the pyloric sphincter into the **duodenum** (the first part of the small intestine), its acidity poses a risk to the delicate intestinal lining.

Specialised **enteroendocrine S cells**, located in the mucosa of the duodenum and upper jejunum, are highly sensitive to this drop in pH. When the pH falls **below approximately 4.5**, these cells are activated and secrete the hormone **secretin** directly into the **bloodstream**.

Secretin then travels through the circulatory system to target organs — primarily the **pancreas and liver** — initiating processes that neutralise the acidic chyme and prepare it for enzymatic digestion in the small intestine.

2. Effects of Secretin on Digestive Secretions

Once released into the bloodstream, **secretin** acts as a chemical messenger, travelling to its target organs to coordinate the neutralisation of acidic chyme and optimise conditions for digestion.

a) Pancreatic Juice Secretion

Secretin binds to receptors on the **duct cells of the pancreas** (distinct from the enzyme-secreting acinar cells). This stimulation causes the duct cells to release a **clear, watery, bicarbonate-rich pancreatic juice**.

- **Bicarbonate ions (HCO_3^-)** combine with hydrogen ions from the chyme, effectively **neutralising gastric acid.**
- The resulting environment raises the duodenal pH to approximately **7.8–8.4**, the optimal range for the activity of **pancreatic enzymes** (amylase, lipase, proteases) and **intestinal brush-border enzymes**.
- This alkaline buffer also protects the **mucosal lining** of the duodenum from acid-induced injury.

b) Bile Secretion

Secretin also targets the **bile ducts within the liver** (and to some extent, the epithelial lining of the gallbladder).

- It stimulates these ducts to secrete a **bicarbonate-rich watery fluid** into the bile.
- This action **increases both the volume and alkalinity of bile**, enhancing its ability to emulsify dietary fats when released into the duodenum.
- The bicarbonate component further contributes to **acid neutralisation**, ensuring that the duodenal environment remains suitable for lipid-digesting enzymes such as pancreatic lipase.

3. Functional Significance

The coordinated action of **secretin** on the pancreas and liver is vital for maintaining both **digestive efficiency** and **intestinal health**.

- **Protection of Intestinal Mucosa:**

  By triggering the release of **bicarbonate-rich fluids**, secretin prevents the **corrosive action of gastric acid** on the delicate epithelial lining of the duodenum. This reduces the risk of acid-induced inflammation or ulceration.

- **Optimisation of Enzyme Activity:**

  Digestive enzymes such as **pancreatic lipase, amylase, and proteases** require a **slightly alkaline environment** to function effectively. Secretin ensures that the pH in the duodenum is consistently within the **optimal range**, allowing for complete breakdown of carbohydrates, lipids, and proteins.

- **Enhanced Fat Digestion:**
  By maintaining an alkaline pH, secretin ensures that **bile salts**—which are essential for emulsifying fats—remain **chemically stable and active**. This greatly improves the efficiency of lipid digestion and absorption in the small intestine.

▶ Role of Secretin

Stimulus	Target Organ	Secretion Triggered	Main Purpose
Acidic chyme in duodenum	Pancreas	Bicarbonate-rich pancreatic juice	Neutralises acid, protects mucosa
Acidic chyme in duodenum	Liver (via bile ducts)	Bicarbonate-rich bile	Maintains alkaline pH for fat digestion

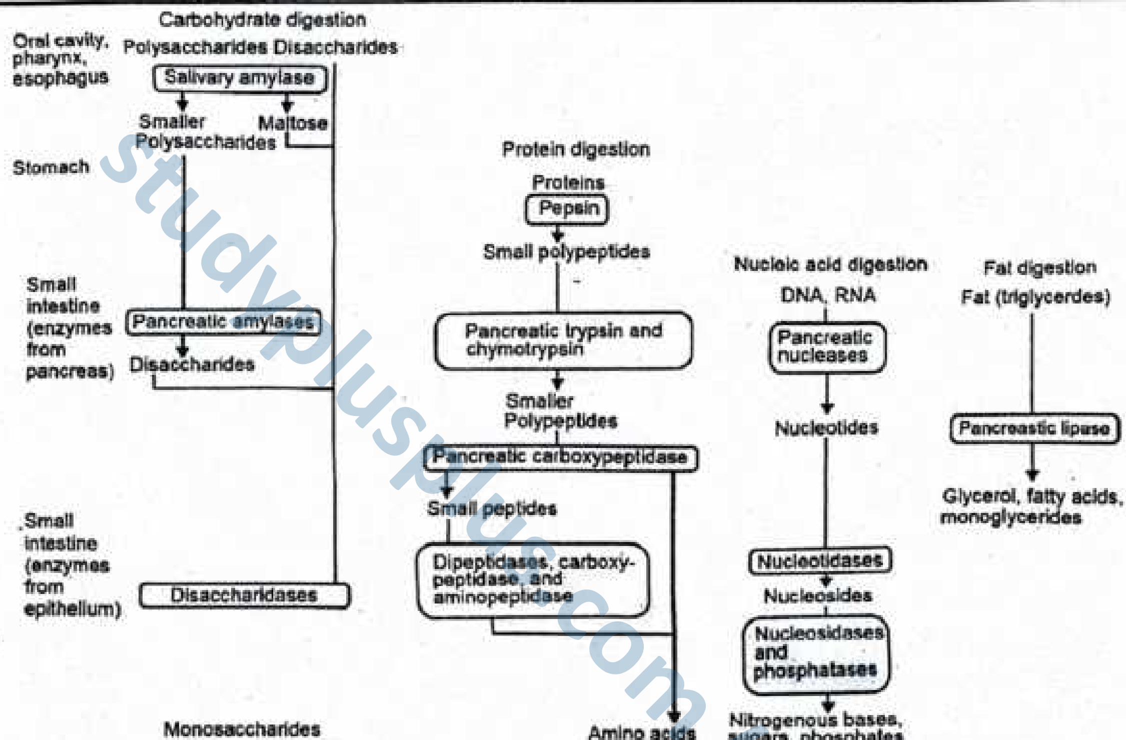

Figure: Flow chart showing action of enzymes in GIT from mouth to small intestine

▶ Compare Bile and Pancreatic Juice and the Hormone Secretin

Feature	Bile	Pancreatic Juice	Secretin (Hormone)
Source	Produced by the liver; stored in the gall bladder	Secreted by the exocrine pancreas	Secreted by endocrine cells in the duodenal wall
Nature	Alkaline, yellow-green fluid	Clear, alkaline fluid containing digestive enzymes and bicarbonate	Peptide hormone
Main Components	Bile salts, bile pigments (e.g. bilirubin), cholesterol, bicarbonate	Digestive enzymes (amylase, lipase, proteases), bicarbonate ions	No digestive components; functions as a chemical messenger
pH	Alkaline (pH ~8)	Alkaline (pH ~8.0–8.5)	Not applicable
Function in Digestion	Emulsifies fats, increasing surface area for lipase action	Breaks down carbohydrates, proteins, and lipids	Stimulates secretion of bicarbonate-rich fluids from liver and pancreas
Enzymes Present	None	Amylase, trypsinogen, chymotrypsinogen, lipase, nucleases	None

Role in pH Regulation	Neutralises acidic chyme via bicarbonate	Neutralises chyme and provides optimal pH for enzyme action	Indirectly regulates pH by promoting bicarbonate secretion
Stimulus for Release	Presence of fats in duodenum (via CCK)	Presence of acidic chyme and nutrients in duodenum (via secretin and CCK)	Presence of acidic chyme in the duodenum
Mode of Transport	Through bile duct into duodenum	Through pancreatic duct into duodenum	Transported via bloodstream to target organs (liver and pancreas)

SLOs Based (MCQs)

1. Which of the following enzymes is secreted by the pancreas in an inactive form and activated in the duodenum?

A. Amylase B. Lipase
C. Trypsinogen D. Maltase

2. What is the main role of secretin in the regulation of pancreatic juice secretion?

A. Stimulates bile release from gall bladder
B. Stimulates bicarbonate-rich fluid from pancreatic duct cells
C. Activates trypsinogen into trypsin
D. Enhances absorption in the small intestine

3. The endocrine portion of the pancreas is responsible for:

A. Producing lipases
B. Neutralizing chyme
C. Digesting nucleic acids
D. Regulating blood glucose

Answer Key

1. C	2. B	3. D

Short Questions

Q.1 Explain the dual role of the pancreas in digestion and hormonal regulation.

Ans. The pancreas performs both **exocrine** and **endocrine** functions. The **exocrine** role involves secreting pancreatic juice into the duodenum via the pancreatic duct. This juice contains digestive enzymes—amylase, lipase, and proteases—to break down carbohydrates, fats, and proteins, respectively. It also includes bicarbonate ions that neutralize acidic chyme from the stomach, creating an optimal pH for enzyme action. The **endocrine** role is carried out by the Islets of Langerhans. Alpha cells secrete glucagon, which raises blood glucose by stimulating glycogen breakdown and gluconeogenesis. Beta cells secrete insulin, which lowers blood glucose by promoting glucose uptake and storage as glycogen. Together, these endocrine hormones maintain glucose homeostasis, while the exocrine products ensure efficient digestion.

> **MARKING SCHEME (3 Marks)**
> - Endocrine role (hormones + effect) – (1.5)
> - Exocrine role (enzymes + bicarbonate + function) – (1.5)

Q.2 Justify why pancreatic proteases are secreted in an inactive form.

Ans. Pancreatic proteases, such as trypsinogen, chymotrypsinogen, and procarboxypeptidase, are secreted as **inactive zymogens** to prevent **autodigestion** of the pancreatic tissue. If secreted in their active form, these powerful proteolytic enzymes would begin digesting the proteins of the pancreas itself, leading to tissue damage and potentially severe conditions like acute pancreatitis. Activation occurs only after the enzymes enter the duodenum. Here, the intestinal enzyme enterokinase converts trypsinogen into trypsin. Trypsin then activates other protease precursors, ensuring protein digestion happens exclusively within the intestinal lumen, where dietary proteins are targeted. This safety mechanism preserves pancreatic integrity while still enabling efficient protein digestion.

> **MARKING SCHEME (3 Marks)**
> - Risk of autodigestion explained – (1.5)
> - Activation site and process – (1.5)

Q.3 Analyse how secretin and CCK coordinate the secretion of bile and pancreatic juice.

Ans. Secretin and cholecystokinin (CCK) are hormones that work together to regulate bile and pancreatic juice secretion. **Secretin** is released from the duodenal mucosa when acidic chyme enters from the stomach. It stimulates the pancreas to release **bicarbonate-rich fluid**, which

> **MARKING SCHEME (3 Marks)**
> - Role of CCK in bile secretion – (1.5)
> - Role of CCK in pancreatic juice secretion – (1.5)

neutralizes the acidity, and prompts the liver to secrete bile. **CCK** is secreted when partially digested fats and proteins are present in the duodenum. It stimulates the **gallbladder** to contract, releasing stored bile that emulsifies fats, and also triggers the pancreas to release **enzyme-rich juice** containing amylase, lipase, and proteases for further digestion. By acting in sequence—secretin preparing the environment and CCK providing digestive tools—these hormones ensure optimal chemical digestion in the small intestine.

Exercise

I Multiple Choice Questions (MCQs)

❖ **Select the correct Ans.**

1. Which among the following is the longest?
A. stomach B. esophagus
C. small intestine D. large intestine

2. Which structure does the appendix come off?
A. transverse colon B. descending colon
C. ascending colon D. small intestine

3. Which part of the Large intestine attaches to the appendix?
A. transverse colon B. descending colon
C. ascending colon D. cecum

4. What is the name of the part of the stomach attached to the esophagus?
A. fundus B. pylorus
C. body D. cardia

5. Where does the pancreatic duct directly join to?
A. jejunum B. duodenum
C. ileum D. liver

6. Where does the body of the stomach lie between?
A. fundus and pyloric antrum
B. pyloric antrum and cardia
C. fundus and cardia
D. cardia and pyloric antrum

7. Which one is the largest gland in the body?
A. liver B. gallbladder
C. pancreas D. large intestine

8. Pepsinogen is activated to pepsin by
A. active secretin B. hydrochloric acid
C. active pepsin and HCℓ D. gastrin

9. Liver secretes bile into the
A. duodenum B. ileum
C. jejunum D. ascending colon

10. Emulsification of fat will not occur in the *absence of*
A. lipase B. bile pigment
C. bile salt D. pancreatic juice

11. Fatty acids and glycerol are first absorbed by
A. lymph vessel B. villi
C. blood capillaries D. hepatic portal vein

12. The hormone responsible for stimulating secretion of hydrochloric acid by stomach cells is
A. pepsin B. secretin
C. gastrin D. insulin

13. Enzyme trypsinogen is changed to trypsin by
A. gastrin B. enterokinase
C. secretin D. hydrochloric acid

14. Which of the following hormones stimulates the production of pancreatic juice and bicarbonate?
A. insulin and glucagon
B. cholecystokinin and secretin
C. gastrin and insulin
D. glucagon and insulin

15. Secretin and cholecystokinin are secreted in
A. pyloric region B. Ileum
C. duodenum D. esophagus

16. Which of the following is synthesized and stored in the liver cells?
A. galactose B. lactose
C. glycogen D. insulin

17. Which juice secreted by the alimentary canal plays an important role in the digestion of fats?
A. pancreatic juice, saliva
C. mucus, Hydrochloric acid
B. saliva, hydrochloric acid
D. bile juice, pancreatic juice

18. What is the role of mucus secreted by the stomach?
A. to digest protein
C. to kill germs in the food
B. to digest fats
D. to protect lining of the stomach

19. **The majority of the water from the indigestible food is absorbed in the ?**

A. stomach B. foodpipe
C. pancreas D. large intestine

20. **Which hydrolytic enzymes react in a low pH environment?**

A. peroxidases B. hydrolases
C. amylases D. proteases

Answer Key

1. C	2. C	3. D	4. D	5. B	6. A	7. A	8. C	9. A	10. C
11. A	12. C	13. B	14. B	15. C	16. C	17. D	18. D	19. D	20. D

II Short Answer Questions

Q.1 Define mechanical digestion.

Ans. Mechanical digestion is the **physical breakdown** of large pieces of food into smaller fragments without altering their chemical composition. It increases the surface area available for enzymes to act during chemical digestion. In humans, it begins in the mouth, where the teeth cut, tear, and grind food during **mastication**. The tongue helps position the food for chewing and forms it into a bolus for swallowing. In the stomach, **churning** mixes the food with gastric juices, further breaking it down into semi-liquid chyme. Mechanical digestion also includes segmental contractions in the small intestine that mix chyme with digestive secretions. This process ensures that food particles are small enough and well-exposed to enzymes for efficient nutrient breakdown and absorption.

MARKING SCHEME (3 Marks)
- Definition of mechanical digestion – 1
- Examples in mouth and stomach – 1
- Function (surface area for enzymes) – 1

Q.2 Describe chemical digestion.

Ans. Chemical digestion refers to the **enzymatic breakdown** of complex food molecules into smaller, absorbable units. This process changes the chemical structure of the food, unlike mechanical digestion. It begins in the mouth, where salivary amylase starts breaking down starch into maltose. In the stomach, pepsin hydrolyses proteins into smaller polypeptides under acidic conditions. Most chemical digestion occurs in the small intestine. Pancreatic enzymes such as amylase, lipase, and proteases act on carbohydrates, fats, and proteins, respectively. Brush border enzymes in the intestinal lining, such as maltase, lactase, and peptidases, complete the breakdown into glucose, amino acids, and fatty acids. These smaller molecules can then be absorbed into the bloodstream or lymph for distribution to body cells.

MARKING SCHEME (3 Marks)
- Definition of chemical digestion – 1
- Enzyme examples with substrates – 1
- End products – 1

Q.3 Explain peristalsis.

Ans. Peristalsis is a **wave-like, rhythmic contraction** and relaxation of smooth muscles in the walls of the digestive tract that propels food and digestive contents forward. It is an involuntary process controlled by the autonomic nervous system. Peristalsis begins in the oesophagus after swallowing, pushing the bolus towards the stomach. In the stomach, it helps mix food with gastric juices. In the small intestine, peristaltic waves move chyme along while segmentations mix it with enzymes and bile for digestion and absorption. In the large intestine, peristalsis moves fecal matter towards the rectum for elimination. This coordinated movement ensures that food passes at an optimal speed for digestion and absorption without backflow.

MARKING SCHEME (3 Marks)
- Definition of peristalsis – 1
- Locations in digestive tract – 1
- Function in movement of food – 1

Q.4 Identify and state the function of epithelial cells in the human stomach.

Ans. The stomach lining contains specialized **epithelial cells** that secrete substances essential for digestion and protection. **Parietal cells** secrete **hydrochloric acid (HCl)**, *which creates* an acidic environment, activates *pepsinogen into* pepsin, and kills ingested microbes. **Chief cells** secrete **pepsinogen**, an inactive enzyme precursor that is converted to pepsin for protein digestion. **Mucous cells** secrete **mucus**, which coats the stomach lining to protect it from the corrosive effects of acid and digestive enzymes. Together, these epithelial cells maintain the stomach's

MARKING SCHEME (3 Marks)
- Parietal cells + function – 1
- Chief cells + function – 1
- Mucous cells + function – 1

digestive function while preventing self-digestion. Their secretions work in coordination, ensuring efficient breakdown of proteins and safeguarding the gastric wall.

Q.5 Give one reason why some enzymes in the stomach and intestine are secreted in inactive forms.

Ans. Certain digestive enzymes, such as **pepsinogen** in the stomach and **trypsinogen** in the pancreas, are secreted as **inactive precursors (zymogens)** to protect the tissues of the organs where they are produced. If released in their active form, these enzymes would begin digesting the structural proteins of the glandular cells and surrounding tissues, leading to self-digestion and possible tissue damage. Activation occurs only when the enzymes reach their target location. For example, pepsinogen is converted to pepsin by stomach acid, and trypsinogen is activated by enterokinase in the duodenum. This mechanism ensures that powerful proteolytic enzymes function only where and when they are needed.

MARKING SCHEME (3 Marks)
- Risk of self-digestion explained – 1.5
- Example with activation process – 1.5

Q.6 Identify the main enzymes involved in protein digestion and state their sources.

Ans. Protein digestion involves several enzymes secreted at different stages of the digestive tract. In the stomach, *pepsin*, derived from inactive pepsinogen, breaks proteins into smaller peptides under acidic conditions. In the small intestine, pancreatic enzymes such as *trypsin* and *chymotrypsin* further hydrolyse peptide bonds, producing smaller peptides. These enzymes are secreted in inactive forms (*trypsinogen*, *chymotrypsinogen*) and activated in the duodenum. The final stage involves *peptidases* (also called proteases) produced by the intestinal lining, which break down peptides into amino acids for absorption. This sequential action ensures complete protein digestion while preventing autodigestion of tissues.

MARKING SCHEME (3 Marks)
- Stomach enzyme and role – 1
- Pancreatic enzymes and role – 1
- Intestinal enzymes and role – 1

Q.7 Predict the effects on digestion if hydrochloric acid is absent from the stomach.

Ans. Hydrochloric acid (HCl) is essential for protein digestion in the stomach. Without HCl, *pepsinogen* cannot be converted into active pepsin, greatly impairing the breakdown of proteins. The absence of acidic pH would also reduce the stomach's ability to kill ingested microbes, increasing infection risk. Furthermore, HCl creates the optimal acidic environment for gastric enzymes, and without it, enzyme activity would be significantly reduced. The lack of acidity would also delay the stimulation of hormones like *secretin*, which regulate pancreatic secretions. This disruption would have a downstream effect on digestion in the small intestine.

MARKING SCHEME (3 Marks)
- Pepsin activation impaired – 1
- Reduced microbial killing – 1
- Loss of optimal enzyme pH – 1

Q.8 Explain how the stomach prevents damage from its own acid and enzymes.

Ans. The stomach is lined with specialised mucous cells that secrete a thick, alkaline mucus layer. This mucus coats the stomach epithelium, acting as a physical and chemical barrier against hydrochloric acid and digestive enzymes like pepsin. Additionally, epithelial cells produce bicarbonate ions that neutralise any acid penetrating the mucus. Rapid regeneration of epithelial cells (every few days) also contributes to maintaining the stomach's integrity. Together, these protective mechanisms prevent acid-induced ulceration and allow the stomach to carry out digestion without self-damage.

MARKING SCHEME (3 Marks)
- Role of mucus barrier – 1
- Bicarbonate protection – 1
- Rapid cell regeneration – 1

Q.9 Justify why villi are present in the intestine but absent in the stomach.

Ans. Villi are finger-like projections in the lining of the small intestine, designed to increase surface area for nutrient absorption. Each villus contains blood capillaries and a lacteal for transporting absorbed nutrients. The stomach's main function is mechanical and chemical digestion, *not absorption* (except for small amounts of alcohol, water, and certain drugs). Therefore, villi are unnecessary in the stomach. Instead, the stomach has gastric pits to maximise secretion of acid, enzymes, and mucus, optimising its digestive role.

MARKING SCHEME (3 Marks)
- Function of villi – 1
- Presence in intestine for absorption – 1
- Absence in stomach due to digestive role – 1

Q.10 State the source of alkalinity that enables trypsin activity in the small intestine.

Ans. Trypsin functions optimally in an alkaline environment. This alkalinity is provided by bicarbonate ions secreted by the pancreas into the duodenum. The release of bicarbonate-rich fluid is stimulated by the hormone *secretin*, which is secreted by the intestinal mucosa when acidic chyme enters from the stomach. These bicarbonate ions neutralise the gastric acid, raising the pH to the alkaline range required for trypsin and other pancreatic enzymes to function effectively. This ensures efficient digestion of proteins in the small intestine.

MARKING SCHEME (3 Marks)
- Source of bicarbonate – 1
- Role of secretin – 1
- Function of alkalinity for trypsin – 1

Q.11 Predict the effect on intestinal enzyme activity if the small intestine's pH remained at 2.

Ans. A pH of 2 is highly acidic and unsuitable for intestinal enzymes, which function optimally at alkaline pH. Such low pH would denature enzyme proteins, altering their active sites and making them unable to bind substrates. This would halt most enzymatic reactions in the small intestine, leading to incomplete digestion and reduced nutrient absorption. The condition could also damage intestinal mucosa, further impairing digestive efficiency.

MARKING SCHEME (3 Marks)
- pH too low for intestinal enzymes – 1
- Denaturation / inactivation explained – 1
- Impact on digestion/absorption – 1

Q.12 Compare the absorption pathways of fats and glucose.

Ans. Glucose is absorbed directly into blood capillaries of the intestinal villi via active transport or facilitated diffusion. It enters the hepatic portal vein and is transported to the liver. In contrast, fats are digested into fatty acids and glycerol, which are absorbed into epithelial cells, reassembled into triglycerides, and packaged into chylomicrons. These enter lacteals (lymphatic vessels) before being carried to the bloodstream via the thoracic duct. Thus, glucose absorption is direct to blood, whereas fat absorption is indirect via the lymph.

MARKING SCHEME (3 Marks)
- Glucose absorption into blood – 1
- Fat absorption into lacteals – 1
- Key pathway difference – 1

Q.13 Describe the defaecation reflex in infants.

Ans. In infants, defaecation is an involuntary process controlled entirely by the spinal reflex. Stretching of the rectal wall by accumulated faeces activates stretch receptors, sending impulses to the spinal cord. This triggers relaxation of both the internal and external anal sphincters without conscious control, resulting in automatic passage of faeces. As the nervous system matures, voluntary control develops.

MARKING SCHEME (3 Marks)
- Trigger: rectal wall stretching – 1
- Involuntary sphincter relaxation – 1
- Automatic passage of faeces – 1

Q.14 Describe the defaecation reflex in adults.

Ans. In adults, defaecation involves both involuntary reflex and voluntary control. Rectal wall stretching activates the reflex pathway, causing contraction of rectal muscles and relaxation of the internal anal sphincter. The external anal sphincter, under voluntary control, must be consciously relaxed for defaecation to occur. Adults can delay defaecation by contracting the external sphincter and pelvic floor muscles until a suitable time.

MARKING SCHEME (3 Marks)
- Reflex initiation by rectal stretching – 1
- Internal sphincter relaxation – 1
- Voluntary control via external sphincter – 1

Q.15 Justify the importance of bile in digestion despite it containing no enzymes.

Ans. Bile plays a key role in fat digestion through *emulsification*. It breaks large fat globules into smaller droplets, increasing the surface area for the enzyme lipase to act. This process enhances the efficiency and speed of fat digestion. Bile also helps neutralise acidic chyme, providing an optimal alkaline pH for lipase activity. Without bile, fat digestion and absorption would be greatly reduced.

MARKING SCHEME (3 Marks)
- Emulsification role – 1
- Increased surface area for lipase – 1
- pH neutralisation for enzyme action – 1

Q.16 State the role of the hormone gastrin in digestion.

Ans. Gastrin is secreted by *G cells* in the pyloric region of the stomach. Its primary role is to stimulate the gastric glands to secrete gastric juice, particularly hydrochloric acid (HCl). The presence of HCl creates an acidic environment necessary for the activation of pepsinogen into pepsin, enabling protein digestion. Gastrin also enhances stomach motility, aiding the mechanical mixing of food with digestive juices. This coordinated action ensures efficient breakdown of proteins and smooth passage of chyme into the small intestine.

MARKING SCHEME (3 Marks)
- Stimulates secretion of gastric juice/HCL – 1
- Role in protein digestion – 1
- Increases stomach motility – 1

Q.17 Describe the role of the hormone secretin in digestion.

Ans. Secretin is produced by the mucosa of the duodenum in response to acidic chyme entering from the stomach. It stimulates the pancreas to release bicarbonate-rich pancreatic juice. This bicarbonate neutralises stomach acid, raising the pH to an alkaline level suitable for the activity of pancreatic enzymes such as trypsin and amylase. Secretin also promotes bile secretion from the liver, further assisting digestion and maintaining an optimal environment for nutrient breakdown in the small intestine.

MARKING SCHEME (3 Marks)
- Stimulates pancreatic bicarbonate secretion – 1
- Neutralises acidic chyme/creates alkaline pH – 1
- Supports enzyme activity/bile secretion – 1

Q.18 Explain the storage role of the liver.

Ans. The liver serves as the body's primary storage centre for essential nutrients and minerals. It stores glycogen, which can be converted to glucose to maintain blood sugar levels. It also stores fat-soluble vitamins (A, D, E, K) and vitamin B_{12}, releasing them as needed. In addition, the liver stores minerals such as iron (bound to ferritin) and copper, which are essential for blood formation and enzyme activity. This storage function ensures a steady supply of vital nutrients, helping maintain homeostasis even between meals.

MARKING SCHEME (3 Marks)
- Glycogen storage and blood sugar regulation – 1
- Storage of vitamins – 1
- Storage of minerals – 1

Q.19 Define the gall bladder and state its function.

Ans. The gall bladder is a small, muscular, pear-shaped organ located beneath the liver. It stores and concentrates bile produced by the liver. During digestion, especially after the intake of fatty food, the gall bladder contracts to release bile into the duodenum via the bile duct. Bile emulsifies fats, breaking them into small droplets, which increases the surface area for lipase action and facilitates fat digestion and absorption.

MARKING SCHEME (3 Marks)
- Location and structure – 1
- Stores and concentrates bile – 1
- Role of bile in fat digestion – 1

Q.20 Write the differences between (a) pharynx and larynx (b) pepsinogen and pepsin.

MARKING SCHEME (3 Marks)
- Structural/definition difference – 1
- Location difference – 1
- Function/role difference – 1

Ans. (a) Pharynx and Larynx

Feature	Pharynx	Larynx
Definition	Muscular passage common to both respiratory and digestive systems.	Cartilaginous structure involved in sound production and air passage.
Location	Behind nasal and oral cavities.	Below the pharynx, in front of the esophagus.
Function	Conducts air to the larynx and food to the esophagus.	Produces sound and directs air into the trachea.
Role in Systems	Part of respiratory and digestive systems.	Part of the respiratory system only.

(b) Pepsinogen and Pepsin.

Feature	Pepsinogen	Pepsin
Definition	Inactive precursor (zymogen) of pepsin.	Active proteolytic enzyme.
Source	Secreted by chief cells of the stomach.	Formed by activation of pepsinogen in acidic pH.
Activation	Requires hydrochloric acid ($HC\ell$) for activation.	Active in acidic medium, especially in the stomach.
Function	No digestive activity until activated.	Breaks down proteins into smaller peptides.

III Long Answer Questions

Q.1 Describe the process of swallowing in man.

Ans. Please See [SLO B-12-R-25]

Q.2 Describe the human stomach with diagram.

Ans. Please See [SLO B-12-R-26]

Q.3 Describe the structure of human small intestine.

Ans. Please See [SLO B-12-R-28]

Q.4 Explain the absorption of digested products from the small intestine lumen to the blood capillaries and lacteals of the villi.

Ans. Please See [SLO B-12-R-28]

Q.5 Describe the large intestine of man. What are the functions of large intestine?

Ans. Please See [SLO B-12-R-30]

Q.6 What is bile? Describe the composition of bile. What is the role of constituents of bile?

Ans. Please See [SLO B-12-R-33]

Q.7 How secretion of bile is related to the secretion of hormone secretin?

Ans. Please See [SLO B-12-R-35]

Q.8 Write the functions of liver of man.

Ans. Please See [SLO B-12-R-32]

Q.9 What is the structure of pancreas? Explain the functions of pancreas as an exocrine gland.

Ans. Please See [SLO B-12-R-34]